magex

From Wannier Hamiltonians to spin dynamics, statistical mechanics and micromagnetics

complete mathematical formalism · derivations · validation · benchmarks

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Roadmap

Part I – Exchange from Wannier Hamiltonians
tight-binding model, magnetic force theorem, Ozaki contour, rotational invariance, J, D, J^{ani}, A from one run, Bruno and downfolding, Lloyd pair energies, linear spin waves, MPI, validation

Part II – The atomistic spin model
full Hamiltonian, nanostructures, Néel surface terms, FFT dipoles with sheet tail, textures

Part III – Spin dynamics
sLLG, integrators, quantum thermostat, RPA, STT/SOT, longitudinal and spin–lattice dynamics

Part IV – Statistical mechanics
Metropolis, heat bath, exact dipolar MC, parallel tempering, Wang–Landau, constrained MC, T_C

Part V – Energy landscapes
minimisers on $(S^2)^N$, GNEB, sparse Hessians, magnon eigenmodes, HTST with zero modes

Part VI – Micromagnetics
energy functional, boundary conditions, demag, solvers, torques, AFM/altermagnets, LLB

Part VII – Implementation
multithreaded kernels, MPI, CUDA, benchmarks vs VAMPIRE/UppASD/Spirit, conventions, validation

Colour code used throughout

colour	quantity	symbols
green	Hamiltonian, exchange field	H, H_0, X, p_i, Y
orange	spin–orbit coupling	S, $\xi\,\mathrm{L}\cdot\mathrm{S}$
blue	Green's function, eigenstates	G, Ψ, $\varepsilon_{n\mathrm{k}}$, U_a
red	spin-model parameters	J, D, J^{ani}, A
purple	rotations and directions	e_i, n_i, ϕ, $\mathcal{R}$, θ
teal	Hessians, perturbations	K_{ab}, T_i, W_{ia}, D_i
brown	energies, occupation, contour	Ω, f, μ, z_p, w_p
gold	magnons	$\omega(\mathrm{q})$, b_i, $h(\mathrm{k})$
`slate`	code / MPI	`engine.py`, `Allreduce`

Part I – Exchange from Wannier Hamiltonians: goal and motivation

The target: an extended Heisenberg model

Spin Hamiltonian (TB2J convention)

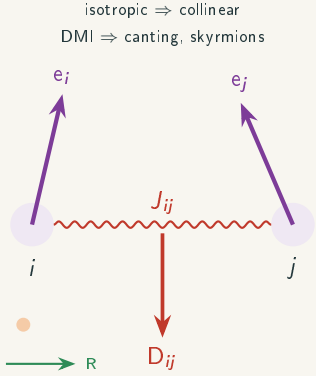
$$E[\{\mathbf{e}\}] = \sum_i \mathbf{e}_i \cdot A_i \cdot \mathbf{e}_i \; - \sum_{i \neq j} \mathbf{e}_i \cdot J_{ij} \cdot \mathbf{e}_j$$

unit spins $|\mathbf{e}_i| = 1$; every ordered pair counted, $J_{ji} = J_{ij}^T$; $J > 0$ ferromagnetic.

Decomposition of the 3×3 tensor

$$J_{ij} = \underbrace{J_{ij}^{\mathrm{iso}} \mathbb{1}}_{\text{Heisenberg}} + \underbrace{J_{ij}^{\mathrm{ani}}}_{\text{sym., traceless}} + \underbrace{J_{ij}^{A}}_{\text{antisym.}}$$

$$-\mathbf{e}_i \cdot J_{ij}^{A} \cdot \mathbf{e}_j = -\mathbf{D}_{ij} \cdot (\mathbf{e}_i \times \mathbf{e}_j), \quad \mathbf{D} = (J_{yz}^{A}, J_{zx}^{A}, J_{xy}^{A})$$



Take-away: every quantity the code prints is a coefficient of this energy function; the job is to extract all of them from one DFT Hamiltonian.

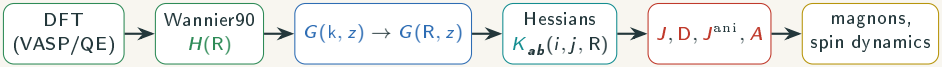
Two ways to get the parameters

Energy mapping

- DFT for many spin configurations, large supercells
- fit J to total energies
- metallisation of high-energy states, supercell-dependent fits, many non-collinear SOC runs

Green's function / force theorem (LKAG)

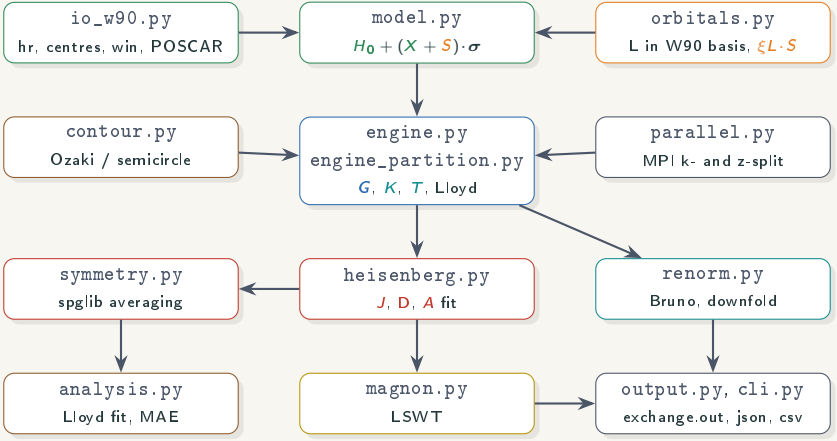
- one ground-state calculation
- J_{ij} at all distances from $G_{ij}(\mathrm{R})$
- local linear response $\Rightarrow$ exact small-angle curvature of the band energy



What limits the standard LKAG/TB2J implementation

limitation	magex
only the *on-site* exchange block is rotated; inter-atomic spin-dependent Wannier elements stay fixed $\Rightarrow$ no global rotation, Goldstone sum rule broken	weighted partition of the full $X(\mathrm{R})$; sum rule exact; bcc Fe J_1: 9.5 $\rightarrow$ 21.2 meV
full tensor needs 3 DFT runs (x, y, z) and a merge	rotate X relative to the lattice in one Hamiltonian
no single-ion anisotropy from Wannier input	A_i from on-site Hessians + band MAE cross-check
$T = 0$ contour with user E_{min}	Ozaki poles at finite T, no E_{min}
plain MFT	Bruno renormalisation
ligand moments rigid/ignored	Schur-complement downfolding
biquadratic from small angles "often unphysical"	exact Lloyd finite-angle pair energies
Wannier noise unquantified	spglib symmetrisation; change reported
magnons isotropic only	LSWT with full J, D, A

Code architecture



driver.py orchestrates; ∼2300 lines Python, numpy/scipy, mpi4py, spglib

I · The tight-binding model

Real space → Bloch

$$H_{mn}(\mathrm{R}) = \langle w_{m0}|\hat{H}|w_{n\mathrm{R}}\rangle \quad\Longrightarrow\quad H(\mathrm{k}) = \sum_{\mathrm{R}} \frac{H(\mathrm{R})}{N_{\mathrm{deg}}(\mathrm{R})}\, e^{2\pi i \mathrm{k}\cdot\mathrm{R}}$$

- `read_hr`: header, Wigner–Seitz degeneracies N_{deg} divided out once.
- `read_win`: `unit_cell_cart`, `atoms_frac`/`cart`, `projections` (s, p, d, f or $l=$), Fortran `1.0d0` floats; `read_poscar` if the .win has no structure.
- Projections are expanded in Wannier90 order: line → site → m; each WF gets (atom, l, m).
- Spinors: `interleaved` (QE, VASP+W90: orbital-major) or `blocked`; internally always spin-block order [↑ all orbitals | ↓ all orbitals].

Collinear input (ISPIN=2)

$$H_0 = \tfrac{1}{2}(H^{\uparrow} + H^{\downarrow}), \qquad X_z = \tfrac{1}{2}(H^{\uparrow} - H^{\downarrow}), \qquad X_{x,y} = S = 0$$

Spin decomposition of a spinor Hamiltonian

Every model is stored as

$$H(\mathrm{R}) = H_0(\mathrm{R}) \otimes \mathbb{1} + \sum_{c=x,y,z} \left[X_c(\mathrm{R}) + S_c(\mathrm{R})\right] \otimes \sigma_c$$

Inverting the Pauli algebra on the 2×2 spin blocks:

$$H_0 = \tfrac{1}{2}(H_{\uparrow\uparrow} + H_{\downarrow\downarrow}), \quad h_z = \tfrac{1}{2}(H_{\uparrow\uparrow} - H_{\downarrow\downarrow}),$$
$$h_x = \tfrac{1}{2}(H_{\uparrow\downarrow} + H_{\downarrow\uparrow}), \quad h_y = \tfrac{i}{2}(H_{\uparrow\downarrow} - H_{\downarrow\uparrow}).$$

since $\sigma_x \to (\uparrow\downarrow, \downarrow\uparrow) = (1, 1)$, $\sigma_y \to (-i, i)$.

	$\uparrow$	$\downarrow$
$\uparrow$	$H_0 + h_z$	$h_x - ih_y$
$\downarrow$	$h_x + ih_y$	$H_0 - h_z$

$\mathrm{h} = X + S$ splits by time-reversal parity (next slide)

Claim

In a time-reversal-symmetric (real) orbital gauge, $X = \mathrm{Re}\,\mathsf{h}$ is *time-odd* (exchange) and $S = i\,\mathrm{Im}\,\mathsf{h}$ is *time-even* (SOC).

Proof

$\Theta = i\sigma_y\mathcal{K}$, $\Theta\sigma_c\Theta^{-1} = -\sigma_c$, and $\mathcal{K}$ conjugates the orbital matrices (real basis functions):

$$\Theta(M \otimes \sigma_c)\Theta^{-1} = -M^* \otimes \sigma_c.$$

Hence the time-even part of h satisfies $M^* = -M$ (purely imaginary) and the time-odd part $M^* = M$ (real). Hermiticity of H gives $\mathsf{h}(\mathrm{R})^\dagger = \mathsf{h}(-\mathrm{R})$, so $\mathrm{Re}\,\mathsf{h}$ is symmetric and $i\,\mathrm{Im}\,\mathsf{h}$ antisymmetric: atomic $\xi\mathrm{L}\cdot\mathrm{S}$ has L imaginary-antisymmetric in a real basis. ■

Diagnostic: $\max|\mathrm{Im}\,H_0|$ is stored; spinor MLWFs are only approximately real, which is why the code also checks $E(\mathrm{n}) = E(-\mathrm{n})$ for the MAE.

Assigning Wannier functions to atoms and to the home cell

- Atom of each WF from the `projections` block (robust), checked against `_centres.xyz`.
- A centre close to an *image* $\boldsymbol{\tau}_a + \mathrm{T}_m$ is moved home: $w'_m(\mathrm{r}) = w_m(\mathrm{r} + \mathrm{T}_m)$.

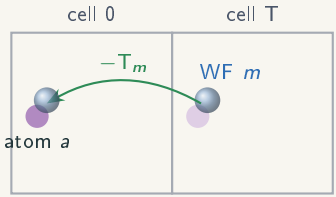
Home-cell shift `_home_shift`

$$H'_{mn}(\mathrm{R}) = H_{mn}(\mathrm{R} + \mathrm{T}_n - \mathrm{T}_m)$$

Proof

$\langle w'_{m\mathbf{0}} | \hat{H} | w'_{n\mathrm{R}} \rangle = \langle w_{m,-\mathrm{T}_m} | \hat{H} | w_{n,\mathrm{R}-\mathrm{T}_n} \rangle$ and translation invariance moves $-\mathrm{T}_m$ to the origin. ■

Residual distance $>$ `wf_centre_tol` (0.6 Å) stops the run: bond-centred MLWFs cannot be attributed to an atom.



Atomic spin–orbit coupling on the Wannier orbitals

Wannier90 real harmonics: $p : (p_z, p_x, p_y)$,
$d : (d_{z^2}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{xy})$, f: 7 functions.

L built from the polynomials themselves

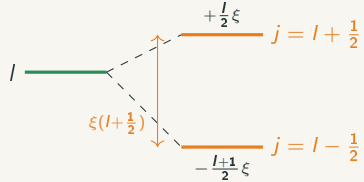
$$L_\alpha \, Y_m = -i(\mathrm{r} \times \nabla)_\alpha \, Y_m = \sum_{m'} Y_{m'} \, C^{\alpha}_{m'm}$$

evaluated on a Gauss–Legendre × uniform-ϕ grid (exact for these polynomials), normalised with exact norms. No tabulated phases ⇒ no sign errors.

Added to the on-site block

$$\xi_l \, \mathrm{L} \cdot \mathrm{S} = \sum_c \underbrace{\frac{\xi_l}{2} L_c}_{S_c} \otimes \sigma_c$$

Checks: $[L_x, L_y] = iL_z$, $\mathrm{L}^2 = l(l{+}1)$, eigenvalues of L·S: $\frac{l}{2}$ and $-\frac{l+1}{2}$.



default ξ table (Cr 35, Fe 55, Pt 600, I 600, Bi 1250 meV); override with `[soc_xi]`.

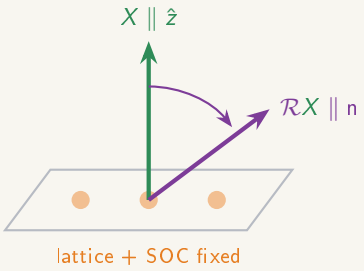
Rotating the magnetisation relative to the lattice

`TBModel.rotated(R)`

$$X_c(\mathrm{R}) \longrightarrow \sum_d \mathcal{R}_{cd}\, X_d(\mathrm{R}), \qquad H_0,\ S \text{ fixed}$$

- rotates the spin direction of the *exchange* field only: the physical rotation of the magnetisation with the crystal and SOC fixed.
- one collinear DFT run + $\xi \mathrm{L}\cdot\mathrm{S}$ gives exact model-level references along x, y, z (or any n).
- a rotation of the *full* spin space would be a gauge change and leave all energies unchanged.

`rotation_to(n)`: Rodrigues rotation taking $\hat{z} \to \mathrm{n}$.



I · Magnetic force theorem

Counting states: the Lloyd formula

Integrated density of states

$$N(\varepsilon) = N_{\mathrm{tot}} - \frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\,\ln\!\left(\varepsilon + i0^{+} - H\right)$$

Proof

For one level E: $\ln(\varepsilon + i0 - E) = \ln|\varepsilon - E| + i\pi\,\theta(E-\varepsilon)$. Summing over eigenvalues, $\frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\,\ln$ counts the levels *above* ε. ■

Variation under $H \to H + V$

$$\delta N(\varepsilon) = -\frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\ln(1 - GV) = \frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\left[GV + \tfrac{1}{2}GVGV + \dots\right], \qquad G = (\varepsilon + i0 - H)^{-1}$$

using $\ln(\varepsilon - H - V) - \ln(\varepsilon - H) = \ln\left(1 - GV\right)$ inside the trace.

Grand potential and the force theorem

Frozen-potential grand potential at fixed

$$\Omega = -k_B T \sum_{n} \ln\!\left(1 + e^{-(\varepsilon_n - \mu)/k_B T}\right) = -\int d\varepsilon\, f(\varepsilon)\, N(\varepsilon)$$

Proof

$\frac{d}{d\varepsilon}\left[- k_B T \ln(1 + e^{-(\varepsilon-\mu)/k_B T})\right] = f(\varepsilon)$; integrate $\Omega = \int n(\varepsilon)\,(\cdots)\,d\varepsilon$ by parts with $n = dN/d\varepsilon$. ■

Second-order energy change

$$\delta\Omega = -\frac{1}{\pi}\,\mathrm{Im}\int d\varepsilon\, f(\varepsilon)\,\mathrm{Tr}\left[G V + \tfrac{1}{2} G V G V\right]$$

Force theorem: to second order the self-consistent energy change equals this band (grand-potential) change at frozen potential, because the first-order density response does not change the functional's stationary value. The code computes Ω-curvatures *exactly*; the only approximation is the frozen potential.

The perturbation: a rigid rotation of one site's exchange field

Rotation by ϕ about axis e_a in spin space: $U = e^{-i\phi\, \mathrm{e}_a \cdot \boldsymbol{\sigma}/2}$.

First derivative

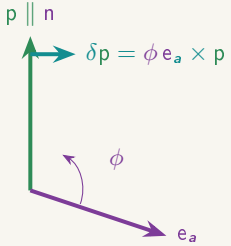
$$\left.\frac{\partial H}{\partial \phi_a}\right|_0 = -\tfrac{i}{2}[\sigma_a, \mathrm{p}\cdot\boldsymbol{\sigma}] = (\mathrm{e}_a \times \mathrm{p})\cdot\boldsymbol{\sigma} \equiv W_a$$

Proof

$[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c \Rightarrow -\tfrac{i}{2}[\sigma_a, p_b\sigma_b] = \epsilon_{abc}\, p_b \sigma_c = (\mathrm{e}_a \times \mathrm{p})_c \sigma_c$. ■

Second derivative (symmetrised)

$$\frac{\partial^2 H}{\partial\phi_a \partial\phi_b} = \tfrac{1}{2}\left[\mathrm{e}_a \times (\mathrm{e}_b \times \mathrm{p}) + \mathrm{e}_b \times (\mathrm{e}_a \times \mathrm{p})\right]\cdot\boldsymbol{\sigma}$$



p_i: time-odd on-site field of site i (`onsite_exchange`). Only $\mathrm{e}_a \perp \mathrm{n}$ gives a non-zero W_a.

Two-site and on-site Hessians magex/engine.py

$$K_{ab}(i,j,\mathrm{R}) \equiv \frac{\partial^2\Omega}{\partial\phi_{ia}\partial\phi_{jb}} = -\frac{1}{\pi}\,\mathrm{Im}\int d\varepsilon\, f(\varepsilon)\,\mathrm{Tr}\left[W_{ia}\,G_{ij}(\mathrm{R})\,W_{jb}\,G_{ji}(-\mathrm{R})\right]$$

$$\frac{\partial^2\Omega}{\partial\phi_{ia}\partial\phi_{ib}} = \tfrac{1}{2}\left[K_{ab}(i,i,0)+K_{ba}(i,i,0)\right] + T_{ab}(i), \qquad T_{ab}(i) = \mathrm{Tr}\left[\rho_{ii}\,\frac{\partial^2 H_i}{\partial\phi_a\partial\phi_b}\right]$$

Why the cross term has no factor $\frac{1}{2}$

With $V = \phi_{ia}W_{ia} + \phi_{jb}W_{jb}$, the term $\frac{1}{2}\,\mathrm{Tr}[G\,V\,G\,V]$ contains $\phi_{ia}\phi_{jb}\,\mathrm{Tr}[G\,W_{ia}\,G\,W_{jb}]$ twice $\times\frac{1}{2}$ (cyclicity). The first-order term with $\partial^2 H$ gives T via $\rho = -\frac{1}{\pi}\,\mathrm{Im}\int f\,G$. ■

$G_{ij}(\mathrm{R})$: block between site i in cell 0 and site j in cell R. All 9 axis pairs (a,b) are computed; those with e ∥ n vanish.

Spectral form (no matrix inversion per z)

$$G(\mathrm{k}, z) = \Psi_{\mathrm{k}}\, \mathrm{diag}\Big(\frac{1}{z - \varepsilon_{n\mathrm{k}}}\Big)\Psi_{\mathrm{k}}^{\dagger}$$

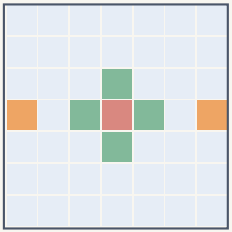
To real space by FFT

$$G(\mathrm{R}, z) = \frac{1}{N_k}\sum_{\mathrm{k}} G(\mathrm{k}, z)\, e^{-2\pi i \mathrm{k}\cdot\mathrm{R}} = \texttt{fftn}/N_k$$

$$G(-\mathrm{R}) = \texttt{roll(flip}(G), 1)$$

- only rows/columns of the spin sites are kept: Ψ_{sel} of shape $(N_k, N_{\mathrm{sel}}, N_b)$
- Γ-centred mesh $\Rightarrow$ R lives on the $N_1 \times N_2 \times N_3$ supercell
- test: K from the k-mesh $=$ Γ-only explicit supercell to 9×10^{-16}

orange: R and R $-$ *N*a tie at the boundary $\Rightarrow$ shared



7×7 supercell of R

I · Energy integration

Continued-fraction expansion

$$f(x) = \tfrac{1}{2} - \sum_{p=1}^{N_p} R_p \left[\tfrac{1}{x - i z_p} + \tfrac{1}{x + i z_p} \right]$$

$z_p = 1/\lambda_p$ from the eigenvalues $\lambda_p > 0$ of the tridiagonal B, $b_n = \frac{1}{2\sqrt{(2n-1)(2n+1)}}$; $R_p = |v_{1p}|^2 / 4\lambda_p^2$.

Contour result

$$-\frac{1}{\pi} \operatorname{Im} \int f \, F(\varepsilon^+) \, d\varepsilon = 2 k_B T \sum_p R_p \operatorname{Re} F(\mu + i z_p k_B T)$$

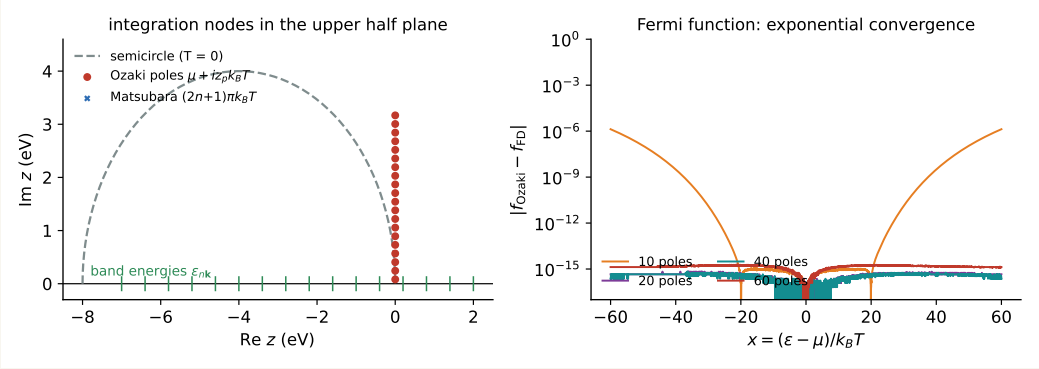
Proof

F analytic in the UHP and $F \sim z^{-2}$: close the real line in the UHP. $\int F/2 = 0$; only the poles of f at $\mu + i z_p k_B T$ (residue $-R_p k_B T$) contribute: $\int fF = -2\pi i \, k_B T \sum_p R_p F(z_p)$. Take $-\frac{1}{\pi}$ Im. ■

Why $F \sim z^{-2}$: every perturbation is a Pauli matrix, $\mathrm{Tr}_\sigma \, \sigma_c = 0$, so the $1/z$ term $\mathrm{Tr}\, V/z$ vanishes.

Gain: no $E_{\min}$, 60 poles for 30 eV at 300 K, $\mu + i\pi k_B T$ smooths k-sums in metals.

Integration nodes and convergence



Left: Ozaki poles sit on the line $\mathrm{Re}\, z = \mu$, sparser than Matsubara frequencies and far from all bands. Right: error of the Fermi function vs. number of poles; `ozaki_npoles` picks the smallest N_p reaching 10^{-10} over the band width. The $T = 0$ semicircle (Gauss–Legendre in angle) is kept as an option.

I · Rotational invariance

The Goldstone sum rule

Exact statement (no SOC)

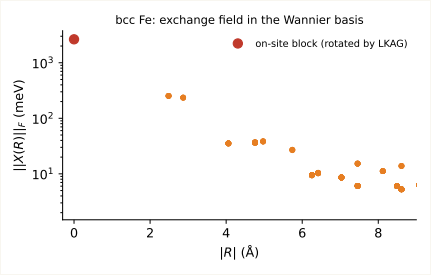
A uniform rotation of *all* spins is a symmetry $\Rightarrow \Omega$ is constant along it:

$$\sum_{j,\mathrm{R}} K_{aa}(i,j,\mathrm{R}) + T_{aa}(i) = 0 \qquad \forall\, i$$

Proof

Rotate every site by the same ϕ: $\Omega(\phi) - \Omega(0) = \frac{1}{2}\phi^2 \sum_{ij\mathrm{R}} \partial^2\Omega = 0$ only if the rotated operators add up to the generator of a *global* rotation, $\sum_i \delta H_i = (\mathrm{e}_a \times X)\cdot\boldsymbol{\sigma}$ for the full $X(\mathrm{R})$. ■

LKAG/TB2J rotate only $\mathrm{p}_i = X_{ii}(0)$. In Wannier bases the exchange field also lives on inter-atomic elements $X_{ij}(\mathrm{R})$, which then never rotate.



Partitioning the exchange field among atoms

Weighted partition

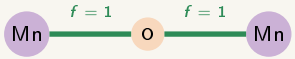
$$X = \sum_i X^{(i)}, \qquad X^{(i)} = \Pi_i X_L + X_L^\dagger \Pi_i$$

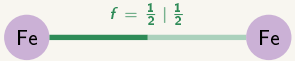
$$(X_L)_{mn}(\mathrm{R}) = f_{a(m)a(n)}\, X_{mn}(\mathrm{R}), \qquad f_{ab} = \frac{w_a}{w_a + w_b}$$

$w_a = \|\mathrm{p}_a\|$; $f_{aa} = \frac{1}{2}$. Π_i projects on the orbitals of atom i in cell 0.

$\sum_i X^{(i)} = X$

$\sum_i \Pi_i = 1$ over all atoms and cells, so $\sum_i X^{(i)} = X_L + X_L^\dagger$. Element-wise $(X_L + X_L^\dagger)_{mn} = (f_{ab} + f_{ba}) X_{mn} = X_{mn}$ using $X^\dagger = X$. ■





magnetic atoms carry the spin-dependent hopping to unpolarised neighbours

Take-away: rotating every $X^{(i)}$ by the same angle is an exact global spin rotation, so the sum rule holds identically.

Perturbation of site i

$$W_{ia} = \Pi_i Y_a + Y_a{}^\dagger \Pi_i, \qquad Y_a = (\mathrm{e}_a \times X_L)\cdot\boldsymbol{\sigma}$$

Trace (`engine_partition`)

$$\begin{aligned}\mathrm{Tr}[W_{ia} G W_{jb} G] = {} & \mathrm{Tr}\left[(Y_a G)_{ij}(Y_b G)_{ji}\right] + \mathrm{Tr}\left[(Y_a G Y_b{}^\dagger)_{ij} G_{ji}\right] \\ & + \mathrm{Tr}\left[G_{ij}(Y_b G Y_a{}^\dagger)_{ji}\right] + \mathrm{Tr}\left[(G Y_b{}^\dagger)_{ij}(G Y_a{}^\dagger)_{ji}\right]\end{aligned}$$

Proof

Expand the product of two sums; move each Π by cyclicity, e.g. $\mathrm{Tr}[Y_a{}^\dagger \Pi_i G Y_b{}^\dagger \Pi_j G] = \mathrm{Tr}[\Pi_i (G Y_b{}^\dagger)\Pi_j (G Y_a{}^\dagger)]$; $\Pi_i \cdots \Pi_j$ selects the (i,j) block. ■

The on-site first-order term becomes $T_{ab}(i) = \mathrm{Tr}_i[(Z_{ab}\rho)_{ii} + (\rho Z_{ab}{}^\dagger)_{ii}]$, $Z_{ab} = \frac{1}{2}[\mathrm{e}_a \times (\mathrm{e}_b \times X_L) + (a \leftrightarrow b)]\cdot\boldsymbol{\sigma}$.

Evaluating the four terms efficiently

Precompute once per k (independent of z):

$$U_a(\mathrm{k}) = Y_a(\mathrm{k})[\mathrm{sel},:]\,\Psi_{\mathrm{k}}, \qquad Q_{ab}(\mathrm{k}) = Z_{ab}(\mathrm{k})[\mathrm{sel},:]\,\Psi_{\mathrm{k}}$$

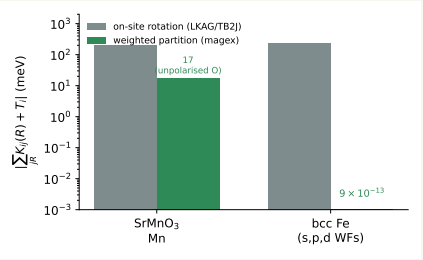
Then every block has the form $A\,\mathrm{diag}(g)\,B^\dagger$ with $g_n = 1/(z-\varepsilon_{n\mathrm{k}})$:

$$G = \Psi_{\mathrm{s}}\, g\, \Psi_{\mathrm{s}}^\dagger, \qquad Y_a\, G = U_a\, g\, \Psi_{\mathrm{s}}^\dagger,$$
$$G\, Y_b^\dagger = \Psi_{\mathrm{s}}\, g\, U_b{}^\dagger, \qquad Y_a\, G\, Y_b^\dagger = U_a\, g\, U_b{}^\dagger$$

16 batched `matmuls` + 16 FFTs per energy node.

Exact checks

- `partition="onsite"` reproduces the fast engine to 4×10^{-16}
- weighted Hessian vs. finite differences of the partitioned rotation: 1.3×10^{-7}
- sum rule: 4×10^{-15}



I · From Hessians to spin-model parameters

Mapping used for every pair `magex/heisenberg.py`

$$K_{ab}(i,j,\mathrm{R}) = -2\,(\mathrm{e}_a \times \mathrm{n}_i)\cdot J_{ij}(\mathrm{R})\cdot(\mathrm{e}_b \times \mathrm{n}_j)$$

Proof

Rotating $\mathrm{e}_i = \mathcal{R}(\phi_i)\mathrm{n}_i$ gives $\delta\mathrm{e}_i = \phi_{ia}\,\mathrm{e}_a \times \mathrm{n}_i + O(\phi^2)$. The pair (i,j) enters E as $-\mathrm{e}_i\cdot J_{ij}\cdot\mathrm{e}_j - \mathrm{e}_j\cdot J_{ji}\cdot\mathrm{e}_i = -2\,\mathrm{e}_i\cdot J_{ij}\cdot\mathrm{e}_j$, whose mixed derivative is $-2(\mathrm{e}_a \times \mathrm{n}_i)\cdot J_{ij}\cdot(\mathrm{e}_b \times \mathrm{n}_j)$. Second-order terms in $\delta\mathrm{e}_i$ multiply n_j and do not contribute to $\partial_{\phi_i}\partial_{\phi_j}$. ■

- collinear reference: $\mathrm{n}_i = s_i\hat{\mathrm{d}}$ with $s_i = \mathrm{sign}(\mathrm{M}_i\cdot\hat{\mathrm{d}})$ (AFM handled automatically)
- isotropic, no SOC: $K_{aa} = -2 s_i s_j J_{ij}$ for $a \perp \hat{\mathrm{d}}$; equivalent to LKAG $J = \frac{1}{4\pi}\,\mathrm{Im}\int \mathrm{Tr}[\Delta_i G_{ij}^{\uparrow}\Delta_j G_{ji}^{\downarrow}]$ averaged over $i \leftrightarrow j$

Three references from one Hamiltonian

Each reference $\hat{\mathrm{d}}$ gives 4 equations (axes $\perp \hat{\mathrm{d}}$):

- $\hat{\mathrm{d}} = \hat{z}$: $J_{xx}, J_{xy}, J_{yx}, J_{yy}$
- $\hat{\mathrm{d}} = \hat{x}$: $J_{yy}, J_{yz}, J_{zy}, J_{zz}$
- $\hat{\mathrm{d}} = \hat{y}$: $J_{zz}, J_{zx}, J_{xz}, J_{xx}$

12 equations, 9 unknowns $\Rightarrow$ least squares:

Joint fit

$$\min_{J_{ij}} \sum_{\hat{\mathrm{d}}} \sum_{a,b} \left| K^{(\hat{\mathrm{d}})}_{ab} + 2(\mathrm{e}_a \times \mathrm{n}_i) \cdot J_{ij} \cdot (\mathrm{e}_b \times \mathrm{n}_j) \right|^2$$

residual = **reference dependence** (printed per pair).

	x	y	z
x	z, y	z	y
y	z	z, x	x
z	y	x	x, y

references that see $J_{\alpha\beta}$
(diagonal: measured twice)

Take-away: one collinear DFT run + $\xi \mathrm{L} \cdot \mathrm{S}$ $\rightarrow$ J^{iso}, J^{ani} and D for all pairs; no three-run merge, no inter-run Wannier noise.

Single-ion anisotropy from the on-site Hessian

Energy of site i alone rotated (all other spins frozen)

$$E_i(\mathrm{e}) = \mathrm{e}\cdot A_i\cdot \mathrm{e} + \mathrm{c}\cdot \mathrm{e}, \qquad \mathrm{c} = -2\sum_{(j,\mathrm{R})\neq(i,0)} J_{ij}(\mathrm{R})\,\mathrm{n}_j$$

Its Hessian (`build_results`)

$$\frac{\partial^2 E_i}{\partial\phi_a\partial\phi_b} = 2(\mathrm{e}_a\times\mathrm{n})\cdot A_i\cdot(\mathrm{e}_b\times\mathrm{n}) + 2\,\mathrm{n}\cdot A_i\cdot \mathrm{S}_{ab} + \mathrm{c}\cdot \mathrm{S}_{ab} + \delta_i\,(\mathrm{e}_a\times\mathrm{n})\cdot(\mathrm{e}_b\times\mathrm{n})$$

$$\mathrm{S}_{ab} = \tfrac{1}{2}\big[\mathrm{e}_a\times(\mathrm{e}_b\times\mathrm{n}) + \mathrm{e}_b\times(\mathrm{e}_a\times\mathrm{n})\big]$$

- equated to $\frac{1}{2}(K_{ab}+K_{ba})(i,i,0)+T_{ab}(i)$ for all three references; unknowns: 5 components of traceless A_i and an isotropic offset δ_i
- images of i in other cells are *different* (frozen) sites: they enter linearly through c
- δ_i absorbs any sum-rule violation; without SOC $A_i = 0$ follows automatically

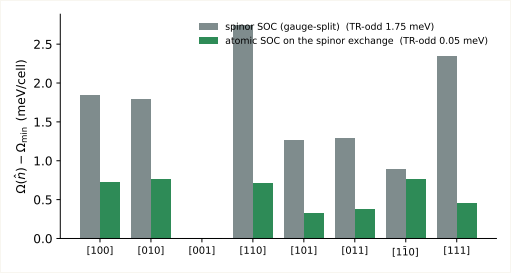
Band MAE (`grand_potential_directions`)

$$\Omega(\mathrm{n}) = -\frac{k_B T}{N_k} \sum_{n\mathbf{k}} \ln\left(1 + e^{-(\varepsilon_{n\mathbf{k}}(\mathrm{n}) - \mu)/k_B T}\right)$$

8 directions and their negatives, dense `mae_kmesh`.

- **time-reversal check:** $\max|\Omega(\mathrm{n}) - \Omega(-\mathrm{n})|/2$ must vanish
- even part fitted to $\mathrm{n} \cdot A_{\mathrm{band}} \cdot \mathrm{n}$
- compared with the Heisenberg prediction

$$E_H(\mathrm{n}) = \sum_i \mathrm{n} \cdot A_i \cdot \mathrm{n} - \sum_{ij\mathrm{R}} s_i s_j \, \mathrm{n} \cdot J_{ij} \cdot \mathrm{n}$$



CrI_3: with gauge-split spinor SOC the TR-odd error is 1.75 meV; with atomic SOC 0.05 meV, easy axis z, $K_{\mathrm{eff}} = 0.71$ (band) vs. 0.78 meV/cell (Heisenberg).

Symmetrisation and supercell aliasing

Group average over the space group (spglib)

$$J(p) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \mathcal{R}_g{}^{T}\, J(g\,p)\, \mathcal{R}_g, \qquad A_i \text{ likewise}$$

Proof

J couples two axial vectors: $\mathrm{e} \to \det(\mathcal{R})\mathcal{R}\mathrm{e}$ for both, $\det^2 = 1$, so $J(gp) = \mathcal{R} J(p) \mathcal{R}^T$; time reversal flips both spins and leaves J unchanged. ■

Atoms are typed by species and |M|; pairs are looked up by their FFT index $\nu = \mathrm{R} \bmod N$; the size of the change is printed as the Wannier-noise estimate.

Minimal images. $\mathrm{R} \equiv \nu \pmod N$ chosen to minimise $|\boldsymbol{\tau}_j + \mathrm{R} - \boldsymbol{\tau}_i|$ (`min_image`). At the supercell boundary several images tie: the coupling is *shared*, J/n_{img} on each (`all_images`), keeping $h(\mathrm{k})$ Hermitian.

$SrMnO_3$: before sharing, $h(\mathrm{k})$ was non-Hermitian and LSWT reported a spurious instability.

I · Beyond the plain force theorem

Bruno renormalisation: rotating moments, not fields

Let θ_i be moment angles and ϕ_i field angles; linear response $\theta = X\phi$.

$$K = K_b + D, \qquad D_i = T_i, \qquad K_b : \text{ bubble incl. on-site}$$

Derivation of $\tilde{K} = -DK_b{}^{-1}D - D$

Hellmann–Feynman: $\partial\Omega/\partial\phi = D(\phi - \theta)$, hence $K = D(1 - X)$, $X = -D^{-1}K_b$. A constraining field fixes the moments at θ while the exchange field follows them; total field $\phi_{\mathrm{tot}} = X^{-1}\theta$. The constraining torque is $\partial E/\partial\theta = D(\phi_{\mathrm{tot}} - \theta) = D(X^{-1} - 1)\theta$, so $\tilde{K} = D(X^{-1} - 1) = -DK_b{}^{-1}D - D$. ■

One FM sublattice

$$\tilde{\omega}(\mathrm{q}) = \frac{\omega(\mathrm{q})}{1 - \omega(\mathrm{q})/\Delta}$$

$\tilde{K} = K + KD^{-1}K + \ldots$ raises spin-wave energies comparable to the exchange splitting Δ.

Implemented per q in the magnetic subspace (`renorm.py`); D from the sum rule so that $\tilde{K}(0)$ keeps the Goldstone zero mode.

Ligand downfolding

Include spin-polarised ligands (O, I, Se . . .) as rotatable sites, then let them relax:

Schur complement per q

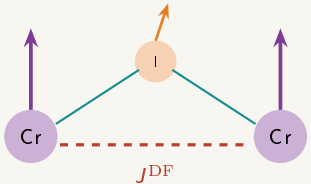
$$K_{\mathrm{eff}}(\mathbf{q}) = K_{MM} - K_{ML} K_{LL}^{-1} K_{LM}$$

Proof

$E = \frac{1}{2}\phi_M^\dagger K_{MM}\phi_M + \mathrm{Re}\,\phi_M^\dagger K_{ML}\phi_L + \frac{1}{2}\phi_L^\dagger K_{LL}\phi_L$;
$\partial E/\partial\phi_L^* = 0$ gives $\phi_L = -K_{LL}^{-1}K_{LM}\phi_M$; substitute. ■

$$J_{ij}^{\mathrm{DF}}(\mathbf{R}) = -\frac{1}{2 s_i s_j}\,\mathrm{FFT}^{-1}\big[K_{\mathrm{eff}}(\mathbf{q})\big]_{ij}$$

Ligands with $\|\mathbf{p}\| < 5$ meV are dropped automatically.



induced ligand moment follows the metals adiabatically

Finite angles: exact Lloyd energies

Local perturbation of a site set S

$$\Delta\Omega_S = \frac{1}{\pi}\,\mathrm{Im}\int d\varepsilon\, f\ \ln\det\left(1 - G_{SS}V_{SS}\right) \;=\; -2k_B T \sum_p R_p\ \ln\left|\det(1 - GV)\right|_{z_p}$$

$V = (\mathcal{R}(\theta)X^{(i)} - X^{(i)})\cdot\boldsymbol{\sigma}$, finite. Only $\ln|\det|$ is needed (Ozaki weights are real): no branch cuts.

Non-local V (weighted partition): Sylvester

$V = \Pi_i A + A^\dagger \Pi_i = LM$, $L = [\Pi_i\ A^\dagger]$, $M = [A; \Pi_i]$. Then $\det(1 - GLM) = \det(1 - MGL)$, a $2n_i \times 2n_i$ matrix with blocks $(AG)_{ii}$, $(AGA^\dagger)_{ii}$, G_{ii}, $(GA^\dagger)_{ii}$; for a pair, the 4×4 block version with phases $e^{\mp 2\pi i \mathbf{k}\cdot\mathbf{R}}$. ■

Pair interaction energy: $E_{\mathrm{int}} = \Delta\Omega_{ij} - \Delta\Omega_i - \Delta\Omega_j$. Exact vs. brute-force Ω: 2.5×10^{-11} eV.

Bilinear and biquadratic parts of the pair energy

Rotate i by $-\theta/2$ and j by $+\theta/2$; $c = \cos(\theta/2)$, $s = s_i s_j$.

Model $E = -\sum_{i \neq j} \left[\quad \mathrm{e}_i \cdot \mathrm{e}_j + \quad (\mathrm{e}_i \cdot \mathrm{e}_j)^2 \right]$

$$E_{\mathrm{int}}(\theta) = 4s\, J'\, c(1-c) - 2B \cos\theta(\cos\theta - 1)$$

$$J_{\mathrm{small}} = J' + 2s\, B$$

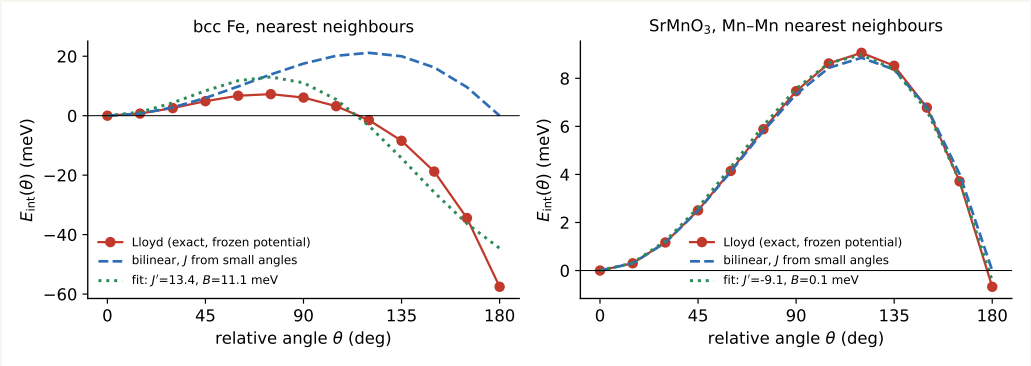
Proof

$E_{\mathrm{int}} = -2J'(\mathrm{e}_i - \mathrm{n}_i)\cdot(\mathrm{e}_j - \mathrm{n}_j) + \ldots$; with $\mathrm{e}_i - \mathrm{n}_i = (-\sin\frac{\theta}{2}, 0, c-1)$, $\mathrm{e}_j - \mathrm{n}_j = s(\sin\frac{\theta}{2}, 0, c-1)$ the dot product is $-2sc(1-c)$. The B term: $(\cos^2\theta - 2c^2 + 1) = \cos\theta(\cos\theta - 1)$. Small θ: $E \simeq \frac{\theta^2}{2}(sJ' + 2B)$. ■



The code also draws the *tensor* bilinear curve $-2(\mathrm{e}_i - \mathrm{n}_i)\cdot J_{ij}\cdot(\mathrm{e}_j - \mathrm{n}_j)$ and reports $\max|E_{\mathrm{Lloyd}} - E_{\mathrm{bil}}|$ as the Heisenberg error.

Lloyd results: how Heisenberg-like are the bonds?



$SrMnO_3$ (localised Mn^{4+}): the bilinear curve from small angles reproduces the exact pair energy over the full range. bcc Fe: strongly angle-dependent nearest-neighbour coupling, a large effective biquadratic term, in line with Szilva *et al.*, PRL 111, 127204 (2013). Small-angle J remain the right input for low-temperature magnons.

I · Spin waves

Local frames: n_i (local z), $\mathsf{u}_i = \mathsf{x}_i + i\mathsf{y}_i$; spins $S_i = |\mathsf{M}_i|/2$; $\tilde{J}_{ij} = -J_{ij}/S_iS_j$, $\tilde{J}_{ii}(0) = A_i/S_i^2$.

Holstein–Primakoff

$$\mathsf{S}_i \simeq \sqrt{S_i/2}\,\left(\bar{\mathsf{u}}_i b_i + \mathsf{u}_i b_i^\dagger\right) + \mathsf{n}_i\left(S_i - b_i^\dagger b_i\right)$$

Bosonic Bogoliubov–de Gennes matrix

$$A_{ij}(\mathsf{k}) = \sqrt{S_iS_j}\;\mathsf{u}_i^T\tilde{J}_{ij}(\mathsf{k})\bar{\mathsf{u}}_j - 2\delta_{ij}\sum_l S_l\;\mathsf{n}_i\cdot\tilde{J}_{il}(0)\cdot\mathsf{n}_l$$

$$B_{ij}(\mathsf{k}) = \sqrt{S_iS_j}\;\mathsf{u}_i^T\tilde{J}_{ij}(\mathsf{k})\mathsf{u}_j, \qquad h(\mathsf{k}) = \begin{pmatrix} A(\mathsf{k}) & B(\mathsf{k}) \\ B(\mathsf{k})^\dagger & A(-\mathsf{k})^T \end{pmatrix}$$

Energies: positive eigenvalues of $\mathrm{diag}(1,-1)\,h(\mathsf{k})$ (paraunitary). Complex eigenvalues $\Rightarrow$ reference not a minimum (warning).

LSWT: checks against closed forms

FM: $\omega(\mathrm{q}) = \frac{4}{M}\big[J(0) - J(\mathrm{q})\big]$

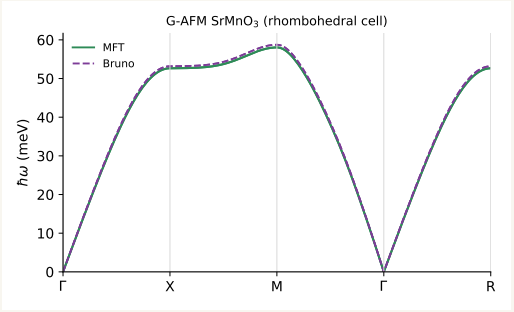
$\mathrm{u}\cdot\bar{\mathrm{u}} = 2$, $\mathrm{u}\cdot\mathrm{u} = 0 \Rightarrow B = 0$ and $A = 2S\big[\tilde{J}(\mathrm{q}) - \tilde{J}(0)\big] = \frac{2}{S}\big[J(0) - J(\mathrm{q})\big]$ with $S = M/2$; the same as the classical precession $\hbar S\dot{\mathrm{e}} = -\mathrm{e} \times \partial E/\partial \mathrm{e}$. ■

AFM chain

$\omega = -\frac{4J}{S}|\sin \pi q|$ reproduced to 10^{-8}.

Easy-axis gap

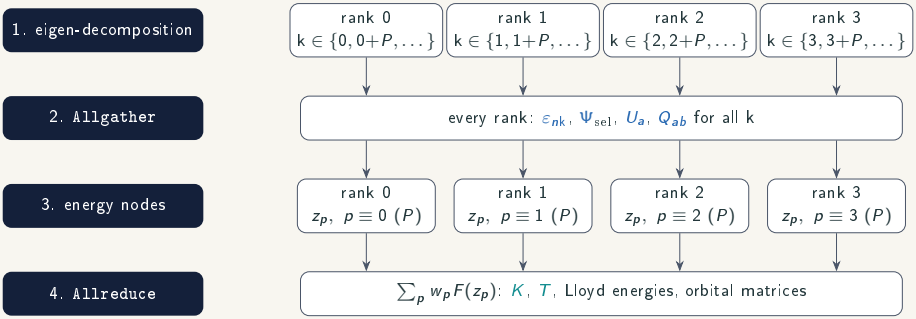
$A = -K\hat{z}\hat{z}$: $\omega(0) = 2K/S$, exact.



G-type $SrMnO_3$: two degenerate Goldstone branches, linear at Γ; Bruno raises the zone-boundary energies slightly (insulator, $\omega \ll \Delta$).

I · Parallel implementation

MPI layout



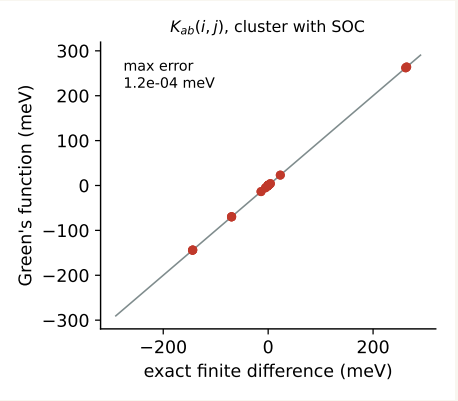
task	cost per energy node	parallel axis
fast engine (on-site)	$N_k N_{\mathrm{sel}}^2 N_b$ + FFT	z
weighted engine	$16 \times N_k N_{\mathrm{sel}}^2 N_b$ + 16 FFTs	z
Lloyd (per pair, angle)	$12 \times N_k n^2 N_b$	z
MAE	N_k diagonalisations per direction	k

$SrMnO_3$, 5^3: 34.6 s → 18.8 s (2 ranks)
results identical to 10^{-14}
`OMP_NUM_THREADS=1`, 1 rank/core

I · Validation and results

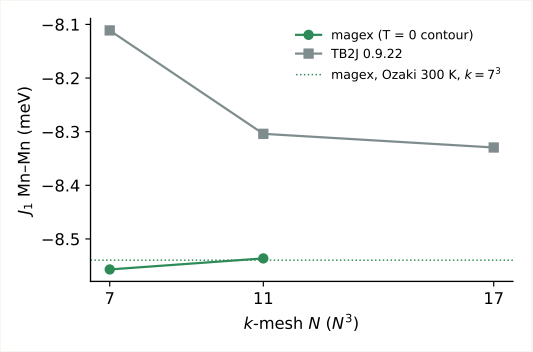
Exactness tests

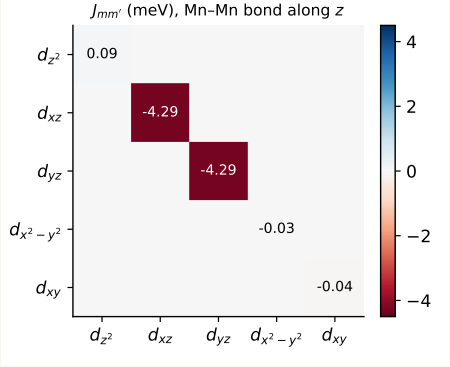
test (`tests/`)	max error
GF Hessian vs. finite differences, cluster with SOC, AFM, spin-dependent hopping	1.5×10^{-7}
weighted partition vs. finite differences	1.3×10^{-7}
k-mesh/FFT vs. explicit supercell	9×10^{-16}
Lloyd vs. exact Ω (both partitions)	2.5×10^{-11}
Goldstone sum rule (weighted)	4×10^{-15}
Heisenberg round trip (J, A)	7×10^{-16}
LSWT: FM, AFM chain, gap	analytic
real $SrMnO_3$: $\sum_R K$ vs. FD	$-66.571 / -66.565$



finite-difference error ($h = 10^{-3}$) dominates the residual.

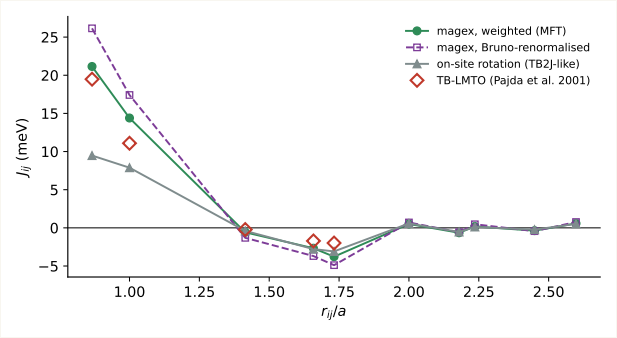
$SrMnO_3$: comparison with TB2J





- same Wannier data, on-site partition: magex $J_1 = -8.54$ meV (converged at 7^3), TB2J 0.9.22 -8.33 meV at 17^3
- t_{2g} channel identical (-4.291 meV for d_{xz}, d_{yz}); the difference sits in the e_g channel
- magex matches brute-force energy differences of the same Hamiltonian to 10^{-4}

bcc Fe: the effect of rotational invariance



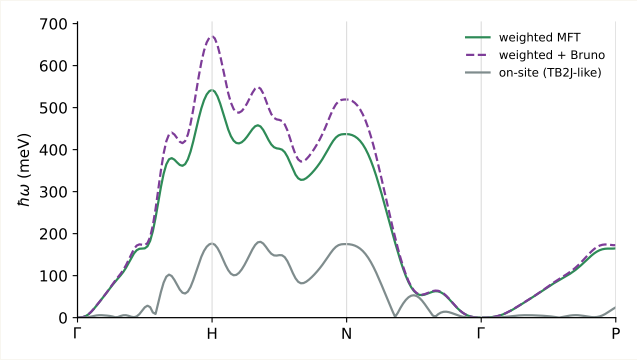
	J_1 (meV)	T_{MF} (K)
on-site (TB2J-like)	9.5	441
weighted, MFT	21.2	1427
weighted, Bruno	26.1	1660
TB-LMTO (Pajda)	19.5	1414
experiment	–	1043

$k_B T_{\mathrm{MF}} = \frac{2}{3} \sum_j J_{0j}$

QE + Wannier90 (s,p,d), 13^3, 2 MPI ranks, 2.7 min.

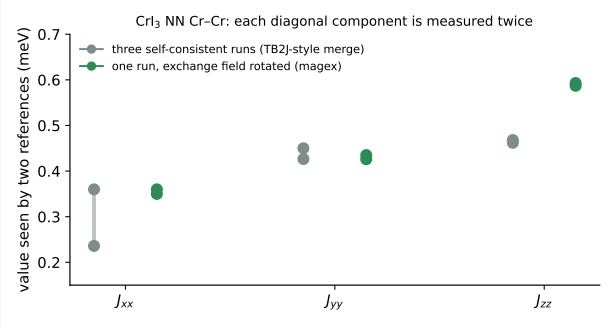
Take-away: rotating the inter-atomic exchange field with the moment turns an unphysical Wannier result into LMTO-quality exchange.

bcc Fe: spin waves



- weighted MFT: maximum 542 meV at H, typical of LKAG Fe results
- Bruno: zone-boundary energies up by $\sim$20% where ω approaches $\Delta \approx 2$ eV
- on-site rotation: soft spectrum, consistent with $T_{\mathrm{MF}} = 441$ K
- the residual wiggles reflect the finite 13^3 real-space range of $J(\mathrm{R})$

CrI_3: one run vs. three runs



- TB2J example data: spinor VASP runs with spins along x, y, z (rotated lattices)
- every diagonal component is measured by two references
- three separately Wannierised runs disagree by up to 35% (J_{xx})
- one run with X rotated agrees with itself to 3%
- TR partner test ($n \to -n$): 1–5%

Take-away: the single-run approach removes inter-run Wannier noise; for spinor input check the TR line, or use collinear + atomic SOC.

I · Using the code

Input and output

```
[input]
win = "wannier90.up.win"
hr_up = "wannier90.up_hr.dat"
hr_dn = "wannier90.dn_hr.dat"
centres = "wannier90.up_centres.xyz"
efermi = 4.327
[magnetic]
elements = ["Mn"];  ligands = ["O"]
[soc_xi]                    # optional atomic SOC
"Mn:d" = 0.045
[calc]
kmesh = [9, 9, 9]
partition = "weighted"    # or "onsite"
refs = "auto"             # x, y, z with SOC
mae = true
[magnon]
path = "GXMGR"
```

```
mpirun -n 40 magex run in.toml
```

exchange.out: moments and charges, sum rule, J (MFT, Bruno, downfolded), D, J^{ani} with residuals, A_i and easy axes, Lloyd table, MAE with TR and Heisenberg checks, orbital-resolved J, timings.

results.json: all pairs in the supercell, full tensors.

pairs.csv, **magnon_band_*.dat/png**.

Limitations: frozen potential; collinear references; spinor input needs a near-real gauge; Bruno/downfolding act on the isotropic channel; no local-axis projections.

Part II – The atomistic spin model

Atomistic Hamiltonian (meV), every ordered pair counted once

$$E[\{\mathrm{e}\}] = \sum_i \mathrm{e}_i \cdot A_i \cdot \mathrm{e}_i - \sum_{i \neq j} \mathrm{e}_i \cdot J_{ij} \cdot \mathrm{e}_j - \sum_{i \neq j} B_{ij} (\mathrm{e}_i \cdot \mathrm{e}_j)^2 - K_4 \sum_{\square} \left[(\mathrm{e}_i \cdot \mathrm{e}_j)(\mathrm{e}_k \cdot \mathrm{e}_l) + (\mathrm{e}_j \cdot \mathrm{e}_k)(\mathrm{e}_l \cdot \mathrm{e}_i) - (\mathrm{e}_i \cdot \mathrm{e}_k)(\mathrm{e}_j \cdot \mathrm{e}_l) \right]$$

$$+ \sum_i \left[K_c \sum_{a<b} \alpha_a^2 \alpha_b^2 - K_2 (\mathrm{e}_i \cdot \mathrm{u})^4 - K_3 (\mathrm{e}_i \cdot \mathrm{u})^6 \right] - \mu_B \sum_i \mu_i \, \mathrm{e}_i \cdot \mathrm{B} - \frac{\mu_B}{2} \sum_i \mu_i \, \mathrm{e}_i \cdot \mathrm{B}_i^{\mathrm{dip}}$$

- $J_{ij} = J_{ij}^{\parallel} + J_{ij}^{S} + J_{ij}^{A}$, with $\mathrm{e}_i \cdot J_{ij}^{A} \cdot \mathrm{e}_j = \mathrm{D}_{ij} \cdot (\mathrm{e}_i \times \mathrm{e}_j)$
- $\alpha_a = \mathrm{e}_i \cdot \mathrm{c}_a$ (cubic axes); $K_c > 0$: easy $\langle 100 \rangle$
- $J > 0$ ferromagnetic; $-J' \sum_{\langle ij \rangle} \Leftrightarrow J = J'/2$

Effective field (tesla), used by every solver:

$$\mathrm{B}_i = -\frac{1}{\mu_B \mu_i} \frac{\partial E}{\partial \mathrm{e}_i}$$

For the bilinear part $\partial E / \partial \mathrm{e}_i = -2 \sum_j J_{ij} \mathrm{e}_j + 2 A_i \mathrm{e}_i$ (using $J_{ji} = J_{ij}^{T}$).

Carving. A boolean mask on the supercell removes sites and all couplings that touch them:

$J \;\to\; P^T J P, \qquad P =$ selection of kept rows/cols (3 per site)

shapes: sphere, cylinder, cube, ellipsoid, truncated octahedron; vacancies; Voronoi roughness.

Materials. Label m_i per site; interface exchange

$J_{ij} \;\to\; f_{m_i m_j} J_{ij} \qquad (f < 0 :$ AF interface, exchange bias)

plus per-material moment, A, K_c, $K_{2,3}$, damping α, temperature.

Néel surface anisotropy (VAMPIRE convention)

$$E_i^s = \frac{k_s}{2} \sum_{j \in nn(i)} (e_i \cdot \hat{r}_{ij})^2$$

only on sites with fewer neighbours than the bulk. It is a quadratic form, folded into the single-ion tensor:

$$A_i \;\leftarrow\; A_i + \frac{k_s}{2} \sum_j \hat{r}_{ij} \hat{r}_{ij}^T$$

so LLG, MC, GNEB and the Hessians see it with no extra code.

Pinned spins: zero torque, zero noise, excluded from MC. Temperature profiles: a site temperature T_i in the noise and in β_i.

Dipole–dipole interaction by FFT

$$\mathsf{B}_i^{\mathrm{dip}} = \frac{\mu_0 \mu_B}{4\pi} \sum_{j \neq i} \mu_j \, \frac{3(\mathsf{e}_j \cdot \hat{\mathsf{r}})\hat{\mathsf{r}} - \mathsf{e}_j}{r^3}, \qquad \mathsf{r} = \mathsf{r}_j - \mathsf{r}_i$$

For sublattices s, t on the supercell grid this is a convolution

$$\mathsf{B}_s(\mathsf{R}) = \sum_{\mathsf{o}} W_{st}(\mathsf{o})\, \mu_t \mathsf{e}_t(\mathsf{R} - \mathsf{o}), \qquad \mathsf{r} = -\mathsf{o} \cdot \mathsf{a} + \mathsf{b}_t - \mathsf{b}_s$$

evaluated with real FFTs (zero padding on open axes).

The sign of o in r matters only when $\mathsf{b}_t \neq \mathsf{b}_s$: the direct-sum test found and fixed this for multi-sublattice systems (error $< 10^{-15}$ now).

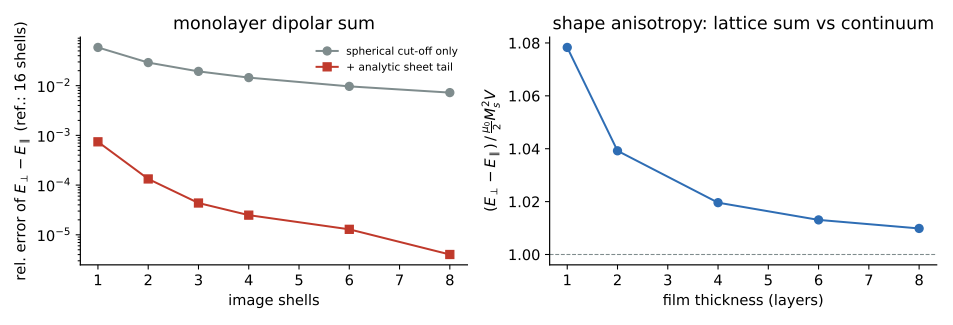
Periodic axes. All image vectors inside a sphere $|\mathsf{r}| \le R_c$ are summed, so each is counted once. For a film the remaining sheets are added analytically:

$$\int_{R_c}^{\infty} 2\pi r \, dr \, \frac{\langle 3\cos^2\varphi - 1 \rangle}{r^3} = \begin{cases} +\pi/R_c & \mathsf{e} \parallel \text{plane} \\ -2\pi/R_c & \mathsf{e} \perp \text{plane} \end{cases}$$

$$W^{\mathrm{tail}} = \frac{\mu_0 \mu_B}{4\pi} \, \frac{\pi}{A_{\mathrm{cell}} R_c} \left(\mathbb{1} - 3\hat{\mathsf{n}}\hat{\mathsf{n}}^{T} \right)$$

relative error 10^{-4} at 2 shells, 4×10^{-6} at 8 (vs 10^{-2} without the tail).

Dipolar sums: convergence and shape anisotropy



Take-away: the lattice sum reproduces the continuum shape anisotropy $\frac{\mu_0}{2}M_s^2V$ for thick films and the correct monolayer value; the sheet tail makes images converge fast.

Initial textures

Skyrmion (radius R, wall w, polarity p, vorticity v, helicity η):

$$\theta(r) = 2\arctan\frac{\sinh(R/w)}{\sinh(r/w)}, \quad \phi = v\,\psi + \eta$$

$$\mathrm{e} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ p\cos\theta)$$

$\eta =$ `"auto"` picks among $0, \pi, \pm\pi/2$ the one with the lowest E. Topological charge $Q = \frac{1}{4\pi}\int \mathrm{e}\cdot(\partial_x \mathrm{e}\times\partial_y \mathrm{e})$: $|Q| = 1$, $Q \to -Q$ for $v = -1$.

Spiral $\mathrm{e} = \cos(\mathrm{q}\cdot\mathrm{r})\hat{\mathrm{n}} + \sin(\mathrm{q}\cdot\mathrm{r})\hat{\mathrm{q}}$ (cycloid) or helix.

Hopfion: stereographic projection and Hopf map, $\mathrm{r}/R \mapsto (z_1, z_2) \in \mathbb{C}^2$,

$$z_1 = \frac{2(x+iy)/R}{1+r^2/R^2},\ z_2 = \frac{2z/R + i(r^2/R^2-1)}{1+r^2/R^2}$$

$$\mathrm{e} = \left(2\,\mathrm{Re}\,z_1\bar{z}_2,\ 2\,\mathrm{Im}\,z_1\bar{z}_2,\ |z_2|^2 - |z_1|^2\right)$$

Domain wall: $\theta(x) = 2\arctan e^{(x-x_0)/w}$, Néel or Bloch.

Part III – Spin dynamics

Stochastic Landau–Lifshitz–Gilbert equation

sLLG, Landau–Lifshitz form

$$\frac{d\,\mathrm{e}_i}{dt} = -\frac{\gamma}{1+\alpha_i^2}\Big[\mathrm{e}_i \times (\mathrm{B}_i + \mathrm{b}_i) + \alpha_i\, \mathrm{e}_i \times \big(\mathrm{e}_i \times (\mathrm{B}_i + \mathrm{b}_i)\big)\Big]$$

Brown's thermal field from the fluctuation–dissipation theorem:

$$\langle b_{ia}(t)\, b_{jb}(t')\rangle = \frac{2\alpha_i k_B T_i}{\gamma \mu_i \mu_B}\, \delta_{ij}\delta_{ab}\delta(t-t')$$

discretised as $\mathrm{b}_i = \sigma_i \boldsymbol{\xi}$, $\sigma_i^2 = 2\alpha_i k_B T_i/(\gamma\mu_i\mu_B\Delta t)$, $\boldsymbol{\xi} \sim \mathcal{N}(0,1)^3$. The noise is multiplicative $\Rightarrow$ Stratonovich calculus; the stationary distribution is $e^{-E/k_B T}$ on $(S^2)^N$.

Units. t in ps, B in T, E in meV, $\gamma = 0.17609$ rad ps^{-1}T^{-1}, $\mu_B = 0.057884$ meV/T.

Per-site α_i (materials), site temperatures T_i (gradients, laser spots), pinned sites ($\mathrm{B}_i = \mathrm{b}_i = 0$).

Write the right-hand side as a tangent vector $\mathsf{f}_i(\mathsf{e})$ and the equivalent rotation vector $\boldsymbol{\omega}_i = \mathsf{e}_i \times \mathsf{f}_i$ (then $\boldsymbol{\omega} \times \mathsf{e} = \mathsf{f}$).

Heun (default), Stratonovich-consistent

$\tilde{\mathsf{e}} = \mathcal{N}[\mathsf{e} + \Delta t\, \mathsf{f}(\mathsf{e}, \mathsf{b})]$,
$\mathsf{e}' = \mathcal{N}\left[\mathsf{e} + \frac{\Delta t}{2}(\mathsf{f}(\mathsf{e}, \mathsf{b}) + \mathsf{f}(\tilde{\mathsf{e}}, \mathsf{b}))\right]$

Depondt–Mertens: Heun on rotations

$\tilde{\mathsf{e}} = \mathcal{R}(\Delta t\, \boldsymbol{\omega}(\mathsf{e}))\, \mathsf{e}, \quad \mathsf{e}' = \mathcal{R}\left(\frac{\Delta t}{2}[\boldsymbol{\omega}(\mathsf{e}) + \boldsymbol{\omega}(\tilde{\mathsf{e}})]\right) \mathsf{e}$
Rodrigues: $\mathcal{R}(\mathsf{w})\mathsf{e} = \mathsf{e}\cos w + (\hat{\mathsf{w}} \times \mathsf{e}) \sin w + \hat{\mathsf{w}}(\hat{\mathsf{w}} \cdot \mathsf{e})(1 - \cos w)$

Semi-implicit midpoint (SIB, Mentink et al.)

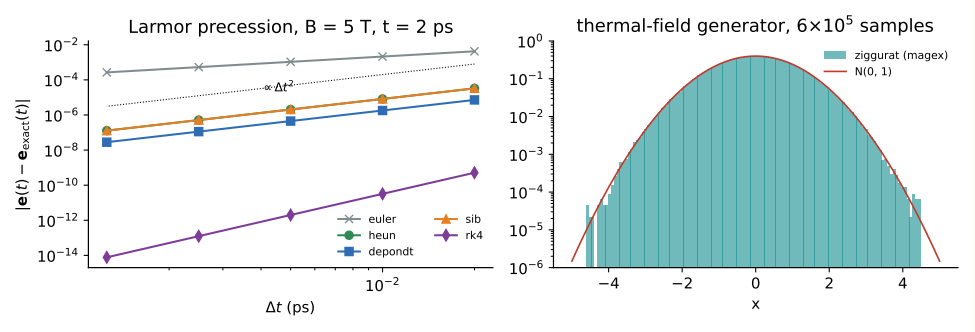
$\mathsf{e}' - \mathsf{e} = \Delta t\, \boldsymbol{\omega}(\bar{\mathsf{e}}) \times \frac{\mathsf{e} + \mathsf{e}'}{2}$ solved exactly by the Cayley transform

$$\mathsf{e}' = \mathsf{e} + \frac{\mathsf{a} \times \mathsf{e} + \frac{1}{2}\mathsf{a} \times (\mathsf{a} \times \mathsf{e})}{1 + |\mathsf{a}|^2/4}, \quad \mathsf{a} = \Delta t\, \boldsymbol{\omega}$$

norm-preserving, time-reversible.

RK4 and Euler are available for deterministic runs.

Integrator accuracy and the thermal-field generator



Take-away: Heun, Depondt and SIB are second order; the ziggurat generator reproduces the Gaussian down to 10^{-6} probability density (tails matter for rare thermal events).

Quantum thermostat

Classical noise populates every magnon with $k_B T$. Replace the white spectrum by Bose statistics (no zero-point term):

$$S(\omega) = 2D\,\frac{x}{e^{x}-1}, \qquad x = \frac{\hbar\omega}{k_B T}$$

represented as a positive mixture of Ornstein–Uhlenbeck processes and damped oscillators,

$$\frac{x}{e^{x}-1} \simeq \sum_{k} w_k \frac{g_k^2}{g_k^2 + x^2} + \sum_{l} v_l \frac{\Omega_l^4}{(\Omega_l^2 - x^2)^2 + \Gamma_l^2 x^2}$$

(NNLS fit, error $\sim 2 \times 10^{-3}$ on $[0, 30]$); every mode integrated exactly over Δt.

RPA (Tyablikov) ordering temperature

$$k_B\, T_{\mathrm{RPA}} = \frac{2}{3}\Big[\frac{1}{nN_q}\sum_{\mathbf{q}} \mathrm{Tr}\big(D - \tilde{J}(\mathbf{q})\big)^{-1}\Big]^{-1}$$

$\tilde{J}_{ij} = s_i s_j J_{ij}$, $D_{ij} = \delta_{ij}\sum_l \tilde{J}_{il}(0)$ + anisotropy gap; quantum spins: $T\,(S+1)/S$.

Spin-transfer and spin–orbit torques on a lattice

Zhang–Li (Gilbert form): $\mathsf{T} = -(\mathsf{u}\cdot\nabla)\mathsf{e} + \beta\,\mathsf{e}\times(\mathsf{u}\cdot\nabla)\mathsf{e}$.
Solving $\dot{\mathsf{e}} = -\gamma\mathsf{e}\times\mathsf{B} + \alpha\mathsf{e}\times\dot{\mathsf{e}} + \mathsf{T}$ for $\dot{\mathsf{e}}$:

$$\dot{\mathsf{e}} = \frac{1}{1+\alpha^2}\big[-\gamma\mathsf{e}\times\mathsf{B} - \gamma\alpha\,\mathsf{e}\times(\mathsf{e}\times\mathsf{B}) + \mathsf{T} + \alpha\,\mathsf{e}\times\mathsf{T}\big]$$

Gradients on any lattice: least squares over the exchange neighbours,

$$(D_a\mathsf{e})_i = \sum_j c_{ij}^a(\mathsf{e}_j - \mathsf{e}_i),\quad c_{ij}^a = w_{ij}\big[M_i^{+}\mathsf{d}_{ij}\big]_a,\ M_i = \sum_j w_{ij}\mathsf{d}_{ij}\mathsf{d}_{ij}^T$$

(sparse, exact for linear fields, works for carved shapes).

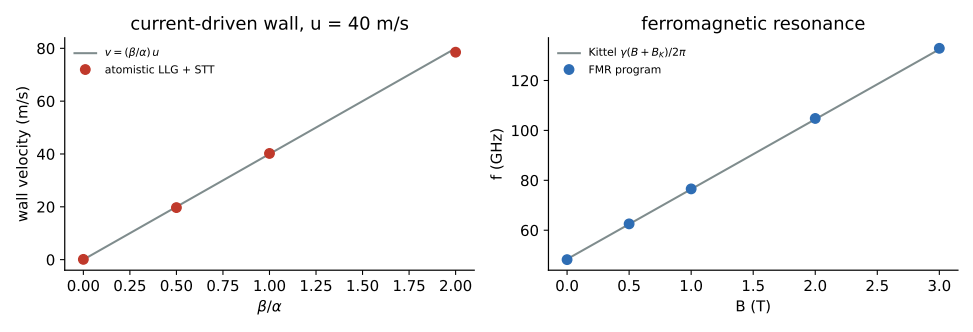
SOT / Slonczewski as fields:
$\mathsf{B} \mathrel{+}= B_{\mathrm{DL}}\,\mathsf{e}\times\mathsf{p} + B_{\mathrm{FL}}\,\mathsf{p}$.

1D wall model, hard-axis field $B_\perp$, below Walker breakdown:

$$v = \frac{\beta}{\alpha}\,u, \qquad u_W = \frac{\alpha\gamma B_\perp \Delta}{2|\alpha-\beta|}$$

Two-temperature model:
$C_e\dot{T}_e = -G(T_e - T_l) + P(t)$,
$C_l\dot{T}_l = G(T_e - T_l)$; the spins see $T_e(t)$.

Torque dynamics and resonance



Take-away: the atomistic Zhang–Li torque gives $v = (\beta/\alpha)u$ to 2%, and the FMR program the Kittel frequency $\gamma(B + B_K)/2\pi$.

Longitudinal spin dynamics

Moments $\mathrm{M}_i = s_i \mathrm{e}_i$ with a Landau term (Ma & Dudarev):

$$E = E_{\mathrm{spin}}[\mathrm{M}] + \sum_i \left(a_i s_i^2 + b_i s_i^4 \right), \quad a_i = \mathrm{h}_i \cdot \mathrm{e}_i^{\mathrm{ref}} - 2 b_i$$

(a_i chosen so that the $T = 0$ reference has $s_i = 1$). Langevin equation:

$$\dot{\mathrm{M}}_i = -\gamma \mathrm{M}_i \times \mathrm{B}_i + \lambda \mathrm{H}_i + \boldsymbol{\xi}_i, \quad \mathrm{H}_i = -\frac{\partial E}{\partial \mathrm{M}_i}$$

$\langle \xi \xi \rangle = 2\lambda k_B T$. Additive noise: Itô = Stratonovich.

Why the ensemble is $e^{-E/k_B T}\, d^3M$

Fokker–Planck: $\partial_t P = -\nabla\cdot(\mathrm{v} P) + \lambda \nabla\cdot(\nabla E\, P + k_B T \nabla P)$ with $\mathrm{v} = -\gamma \mathrm{M} \times \mathrm{B}$. The precession conserves E and is divergence-free, $\nabla\cdot(\mathrm{M} \times \nabla E) = \nabla E \cdot (\nabla \times \mathrm{M}) - \mathrm{M} \cdot (\nabla \times \nabla E) = 0$, so $P \propto e^{-E/k_B T}$ is stationary. ■

Test: $\langle s \rangle$ from LSD and LSF Monte Carlo agree to 10^{-3}.

Coupled spin–lattice dynamics

$$\mathcal{H} = \sum_i \frac{\mathrm{p}_i^2}{2m} + \frac{k}{2}\sum_{\langle ij\rangle}(r_{ij}-r_{ij}^0)^2 + E_{\mathrm{spin}}[\mathrm{e}] - 2\sum_{\langle ij\rangle}\left[J(r_{ij}) - J(r_{ij}^0)\right]\mathrm{e}_i\mathrm{e}_j$$

$J(r) = J_0\, e^{-(r-r_0)/\lambda}$ (or Ma's cubic form) on every isotropic bond. Forces:

$$\mathrm{F}_i = \sum_j \left[k(r_{ij} - r_{ij}^0) - 2J'(r_{ij})\,\mathrm{e}_i\cdot\mathrm{e}_j\right]\hat{\mathrm{r}}_{ij}$$

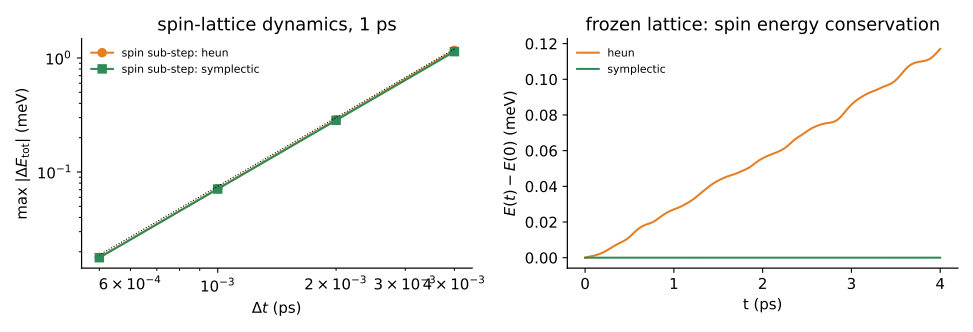
Palindromic splitting (second order)

$$\mathcal{K}_{h/2}\,\mathcal{D}_{h/2}\,[\mathcal{O}_h]\,\mathcal{S}_h\,\mathcal{D}_{h/2}\,\mathcal{K}_{h/2}$$

kick $\mathcal{K}$, drift $\mathcal{D}$, Langevin $\mathcal{O}$, spins $\mathcal{S}$ at the mid-step lattice. $\mathcal{S}_h$: colour groups swept forward then backward with $h/2$, each spin by the implicit midpoint rule (Cayley) $\Rightarrow$ symplectic.

The previous version applied both drifts before the spin step: first order. Fixed.

Spin–lattice dynamics: energy conservation



Take-away: with the palindromic splitting the total energy error falls as Δt^2; the symplectic spin step conserves the spin energy four orders of magnitude better than Heun.

Anharmonic lattice potentials

Bond potentials $V(r)$, $x = r - r_0$, minimum at the reference bond length:

$$V_{\mathrm{harm}} = \tfrac{k}{2}x^2$$

$$V_{\mathrm{FPU}} = \tfrac{k}{2}x^2 + \tfrac{k_3}{3}x^3 + \tfrac{k_4}{4}x^4$$

$$V_{\mathrm{Morse}} = D\left(1 - e^{-ax}\right)^2$$

$$V_{\mathrm{LJ}} = 4\varepsilon\left[\left(\tfrac{\sigma}{r}\right)^{12} - \left(\tfrac{\sigma}{r}\right)^{6}\right] + \varepsilon, \quad \sigma = r_0/2^{1/6}$$

forces $\mathrm{F}_i = \sum_j \left[V'(r_{ij}) - 2J'(r_{ij})\mathrm{e}_i\cdot\mathrm{e}_j\right]\hat{r}_{ij}$.

The cubic term ($k_3 < 0$, Morse) makes the bonds expand on heating: thermal expansion and its coupling to the exchange, $J(r)$, enter the spin dynamics directly (magnetovolume effects, phonon–magnon damping).

Checks: forces $= -\partial\mathcal{H}/\partial\mathrm{u}$ to 10^{-8}; with the palindromic splitting the energy error falls by $4.0\times$ per halving of Δt for every potential.

Part IV – Statistical mechanics

Graph colouring: sites of one colour share no coupling, so a colour is updated in parallel and exactly. Local change for $\mathbf{e}_i \to \mathbf{e}_i'$:

$$\Delta E = -\mathbf{H}_i(\mathbf{e}_i' - \mathbf{e}_i) + \Delta E_i^{\text{on-site}}, \quad \mathbf{H}_i = 2\sum_j J_{ij}\mathbf{e}_j + \mu_B \mu_i \mathbf{B}$$

accepted with $\min(1, e^{-\beta_i \Delta E})$.

Moves: adaptive Gaussian cone (acceptance $\to 50\%$), uniform, Hinzke–Nowak mixture, heat bath; over-relaxation $\mathbf{e}' = 2\frac{\mathbf{e}\cdot\mathbf{H}}{|\mathbf{H}|^2}\mathbf{H} - \mathbf{e}$ (conserves $\mathbf{H}\cdot\mathbf{e}$).

Heat bath for the linear field

Sample $p(\mathbf{e}) \propto e^{\beta \mathbf{H}\cdot\mathbf{e}}$ exactly: with $x = \beta|\mathbf{H}|$, $u \in (0,1)$,

$$\cos\theta = 1 + \frac{1}{x}\ln\left[1 - u\,(1 - e^{-2x})\right], \quad \phi = 2\pi u'$$

then accept with $e^{-\beta\,\Delta E^{\text{on-site}}}$ (independence sampler $\Rightarrow$ detailed balance). Acceptance ≈ 1 for weak anisotropy.

Exact Monte Carlo with long-range dipoles

Simultaneous updates of a colour are not exact when all spins interact. magex keeps the dipolar field and updates it after every accepted single-spin move ($\mathcal{O}(N)$):

$$\mathrm{B}_k^{\mathrm{dip}} \;+=\; W_{s_k s_j}(\mathrm{c}_k - \mathrm{c}_j)\,\mu_j\,\Delta\mathrm{e}_j \quad \forall k$$

Energy change including the coupling of j to its own periodic images W_j^0:

$$\Delta E^{\mathrm{dip}} = -\mu_B \mu_j\,\Delta\mathrm{e}_j\cdot\left(\mathrm{B}_j - \mu_j\,W_j^0\,\mathrm{e}_j\right) - \frac{\mu_B\mu_j^2}{2}\Delta\left(\mathrm{e}_j\cdot W_j^0\,\mathrm{e}_j\right)$$

The field is rebuilt by FFT every sweep (no drift).

Test: periodic film with strong dipoles; brute-force Metropolis on full energies vs the exact kernel: equal $\langle E\rangle$ within 1σ (`tests/test_fast.py`).

Parallel tempering: swap $T_a \leftrightarrow T_b$ with $\min\left(1, e^{(\beta_a-\beta_b)(E_a-E_b)}\right)$.

Wang–Landau: $\ln g(E)$ $+=$ $\ln f$, flat histogram, $1/t$ schedule; $Z(T) = \sum_E g(E)e^{-\beta E}$.

Constrained Monte Carlo and the anisotropy $K(T)$

Fix the direction $\hat{\mathrm{n}}$ of $\mathrm{M} = \sum \mathrm{e}_i$ (Asselin et al. 2010). Pair move: primary i (Gaussian), compensating j restores the transverse components; with $z \parallel \hat{\mathrm{n}}$,

$$e'_{j,xy} = e_{j,xy} + e_{i,xy} - e'_{i,xy}, \quad e'_{jz} = \mathrm{sgn}(e_{jz})\sqrt{1 - e'^2_{jx} - e'^2_{jy}}$$

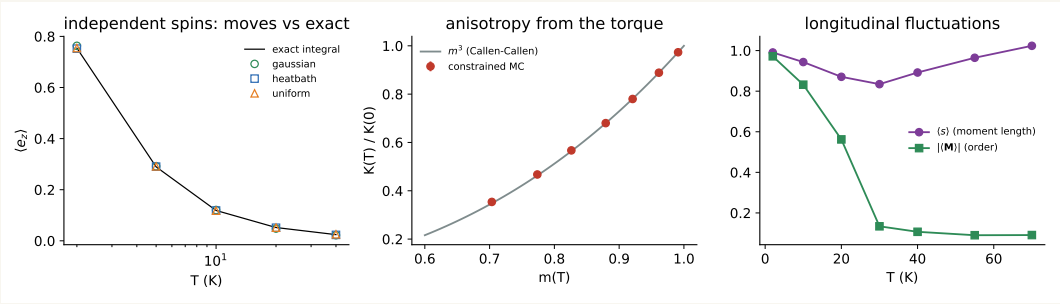
$$P = \min\left[1,\ \left(\frac{M'_z}{M_z}\right)^2 \frac{|e_{jz}|}{|e'_{jz}|}\, e^{-\beta\Delta E}\right]$$

The internal torque gives the free energy: $\langle \boldsymbol{\tau} \rangle \cdot \hat{\boldsymbol{\theta}} = -\partial F / \partial \theta$; uniaxial: $\langle \tau \rangle = -K(T) \sin 2\theta$.

Supported: exchange tensors, DMI, single-ion and surface anisotropy, cubic and higher uniaxial terms, Zeeman, and exact dipoles (incremental field).

Callen–Callen: single-ion $K(T)/K(0) \simeq m^{l(l+1)/2} = m^3$ for $l = 2$. Film with dipoles only: torque at $\theta = 45^\circ$ equals $E_\perp - E_\parallel$ (shape anisotropy) to $< 1\%$.

Monte Carlo checks



Take-away: all move types reproduce exact single-spin averages; constrained MC follows m^3; longitudinal fluctuations keep a finite moment above T_C while the order vanishes.

Ordering temperatures: finite-size scaling

Per spin: E, $C = \frac{\langle E^2\rangle - \langle E\rangle^2}{N k_B T^2}$, order parameter $m = |\sum_i s_i \mathbf{e}_i|/N$ (signs from the magex reference: FM, AFM, ferrimagnet),

$$\chi = \frac{N(\langle m^2\rangle - \langle m\rangle^2)}{k_B T}, \qquad U_4 = 1 - \frac{\langle m^4\rangle}{3\langle m^2\rangle^2}$$

T_C from Binder-cumulant crossings of two sizes; error bars by blocking.

simple-cubic Heisenberg, $J = 1$ meV (magex): exact $T_C = 1.443 \cdot 2J/k_B = 33.49$ K; magex 33.29 K.

bcc Fe (MFT exchange, $r_c = 11$ Å): 1060 K (exp. 1043 K).

The J(q) stability check warns when a truncated coupling set makes a spiral the ground state.

Part V – Energy landscapes

Minimisation on $(S^2)^N$

Tangent gradient $\mathrm{g}_i = (\mathbb{1} - \mathrm{e}_i \mathrm{e}_i^T)\, \partial E / \partial \mathrm{e}_i$; convergence: $\max_i |\mathrm{g}_i| < \epsilon$. Geodesic retraction for a tangent step d_i:

$$\mathrm{e}_i \to \mathrm{e}_i \cos|\mathrm{d}_i| + \hat{\mathrm{d}}_i \sin|\mathrm{d}_i|$$

L-BFGS (Spirit's LBFGS-OSO idea): two-loop recursion on tangent vectors, pairs (s, y) transported by projection, Armijo line search with max angle 0.3 rad.
NCG: Polak–Ribière$^+$ $\beta = \max\left(0, \frac{\mathrm{g}' \cdot (\mathrm{g}' - \mathrm{g})}{\mathrm{g} \cdot \mathrm{g}}\right)$, restarts.
VP: quick-min velocity projection.

When $|\Delta E| < 10^{-11}|E|$ (round-off), steps are accepted if the tangent gradient decreases: torques of 10^{-7} meV are reached for 10^4 spins.

GNEB with climbing image

$\mathrm{F}_\nu = -\nabla E|_\perp + k\,(|\Delta_+| - |\Delta_-|)\,\hat{\boldsymbol{\tau}}_\nu$,
climbing image: $-\nabla E + 2(\nabla E \cdot \hat{\boldsymbol{\tau}})\hat{\boldsymbol{\tau}}$; geodesic distances, upwind tangents, projected velocity Verlet.

Harmonic analysis: Hessian and magnon eigenmodes

Tangent frame $(\mathrm{t}_{i1}, \mathrm{t}_{i2}) \perp \mathrm{e}_i$; Riemannian Hessian

$$H_{(ia),(jb)} = \mathrm{t}_{ia} \cdot \frac{\partial^2 E}{\partial \mathrm{e}_i \partial \mathrm{e}_j} \cdot \mathrm{t}_{jb} - \delta_{ij}\delta_{ab}\, \mathrm{e}_i \cdot \frac{\partial E}{\partial \mathrm{e}_i}$$

Sparse: $-2J + 2A$ + on-site terms analytic, dipolar and higher order matrix-free (gradient differences).

Undamped linearised LLG: $\dot{\mathrm{q}} = GH\mathrm{q}$, $G = \Gamma J$, $\Gamma_i = \gamma/(\mu_i \mu_B)$, $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ per spin.

Frequencies as a symmetric problem

GH has eigenvalues $\pm i\omega$. With $\mathrm{y} = H\mathrm{q}$:

$$(G^T H G)\,\mathrm{y} = \omega^2\, H^{-1}\mathrm{y}$$

symmetric, H^{-1} positive definite at a minimum $\Rightarrow$ shift-invert Lanczos with sparse LU of H.

22 500-spin film: lowest modes in 10 s; $k = 0$ mode equals $\gamma B/2\pi$; FM ring: analytic magnon dispersion to 10^{-8}.

HTST rate (Bessarab, Uzdin & Jónsson 2012)

$$k = \frac{\nu}{2\pi}\sqrt{\frac{\prod' \varepsilon_i^{\min}}{\prod'' \varepsilon_i^{\mathrm{sp}}}}\,(2\pi k_B T)^{\frac{n_0^{\min}-n_0^{\mathrm{sp}}}{2}}\,\frac{V^{\mathrm{sp}}}{V^{\min}}\,e^{-\Delta E/k_B T}, \qquad \nu^2 = \sum_{i\ \text{stable}} \frac{a_i^2}{\varepsilon_i^{\mathrm{sp}}},\quad a_i = \varepsilon_i^{\mathrm{sp}}\,\mathsf{q}_1 \cdot \frac{G}{1+\alpha^2}\mathsf{q}_i$$

$\prod'$, $\prod''$ skip zero modes (and the unstable mode q_1). Translational zero modes: generators $\partial \mathsf{e}/\partial x_a$ from the lattice gradients,

$$V = \Big(\prod_a L_a\Big)\sqrt{\det\left(\partial_a \mathsf{e} \cdot \partial_b \mathsf{e}\right)}$$

Checks: biaxial macrospin: prefactor $= f_{\mathrm{FMR}}$ of the minimum (exact); adding a decoupled free spin with zero modes at both points leaves k unchanged.

Saddle polishing: L-BFGS for the minimum; minimum-mode following (small systems) or Newton steps on the sparse Hessian (large systems) from the climbing image.

HTST without dense diagonalisation

Determinants. With the k lowest modes λ_i from shift-invert Lanczos and a tiny shift s:

$$\ln|\det' H| = \ln|\det(H{+}s)| - \sum_{i\le k,\ \mathrm{drop}} \ln|\lambda_i{+}s| - \sum_{i\le k,\ \mathrm{keep}} \ln\left|1+\frac{s}{\lambda_i}\right|$$

$\ln|\det(H+s)|$ from the diagonal of a fill-reducing sparse LU; neglected remainder $< Ns/\lambda_{\min}$.

Dynamical factor from the unstable mode alone:

$$v^2 = \sum_i \varepsilon_i (\mathrm{q}_1 \cdot G\mathrm{q}_i)^2 = (P\mathrm{w})^T H_{\mathrm{sp}} (P\mathrm{w}), \quad \mathrm{w} = G^T \mathrm{q}_1$$

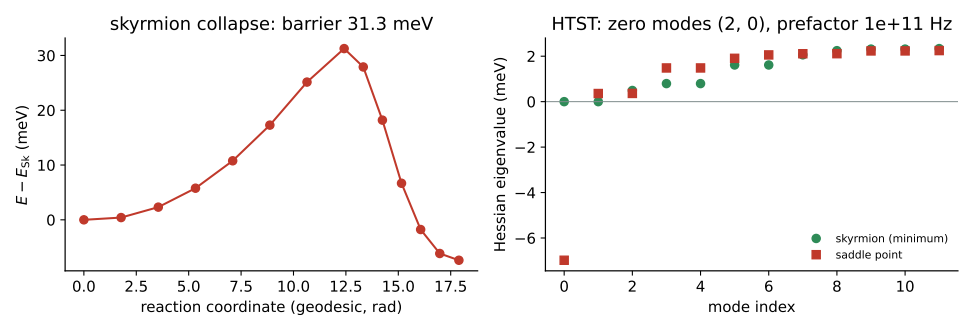
(P removes q_1 and the zero modes).

Saddle multiplicity. A free skyrmion has two translational zero modes, the collapsed saddle is pinned by the lattice: each lattice translation is a separate channel, $k \to N_{\mathrm{cells}}\, k$.

spins	prefactor (Hz)	time
400	3.87×10^{13}	0.1 s
3 600	3.763×10^{13}	0.6 s
14 400	3.763×10^{13}	5.5 s
40 000	3.763×10^{13}	37 s

sparse = dense to 10^{-8}; box-size independent.

Skyrmion collapse: path and harmonic spectra



Take-away: the isolated skyrmion has two translational zero modes; at the collapsed saddle they are lifted by the lattice (Peierls pinning), which HTST accounts for automatically.

Part VI – Micromagnetics

Energy density (J/m^3)

$$
\begin{aligned}
w = {} & A|\nabla \mathrm{m}|^2 + D_{\mathrm{int}}\left[m_z \nabla\cdot\mathrm{m} - (\mathrm{m}\cdot\nabla) m_z\right] + D_{\mathrm{bulk}}\,\mathrm{m}\cdot(\nabla\times\mathrm{m}) - K_{u1}(\mathrm{m}\cdot\mathrm{u})^2 - K_{u2}(\mathrm{m}\cdot\mathrm{u})^4 \\
& + K_{c1}\sum_{a<b}\alpha_a^2\alpha_b^2 + K_{c2}\,\alpha_1^2\alpha_2^2\alpha_3^2 + K_{c3}\Big(\sum_{a<b}\alpha_a^2\alpha_b^2\Big)^2 + B_1\sum_a \epsilon_{aa} m_a^2 + 2B_2\sum_{a<b}\epsilon_{ab} m_a m_b \\
& - M_s \mathrm{m}\cdot\mathrm{B} - \frac{\mu_0}{2} M_s \mathrm{m}\cdot\mathrm{H}_d + w_{\mathrm{custom}}(\mathrm{m}, \mathrm{r}, t)
\end{aligned}
$$

$\mathrm{B}_{\mathrm{eff}} = -\frac{1}{M_s}\frac{\delta w}{\delta \mathrm{m}}$; e.g. magnetoelastic $B_x = -\frac{2}{M_s}\left[B_1\epsilon_{xx} m_x + B_2(\epsilon_{xy} m_y + \epsilon_{xz} m_z)\right]$. Energies from fields: $E_k = -c_k \int M_s \mathrm{m}\cdot\mathrm{B}_k$, $c = \frac{1}{2}$ (quadratic), $\frac{1}{4}$, $\frac{1}{6}$, $\frac{1}{8}$ (quartic, sextic, octic cubic terms).

Custom terms: w_{custom} given as a numpy expression; its field by central differences in m. All terms are tested against finite differences of the discrete energy.

Exchange with link constants (harmonic mean, inter-region scaling):

$$\mathrm{B}_i^{\mathrm{ex}} = \frac{2}{M_{s,i}} \sum_{a,\pm} s_a \frac{A_{i,a}^{\pm}}{d_a^2} (\mathrm{m}_{i\pm a} - \mathrm{m}_i)$$

Free surfaces: ghost cells from the chiral boundary condition

$$2A\,\partial_{\mathrm{n}}\,\mathrm{m} = D_{\mathrm{int}}\left[(\mathrm{m}\cdot\mathrm{n})\hat{z} - m_z\mathrm{n}\right] + D_{\mathrm{bulk}}\,\mathrm{n}\times\mathrm{m}$$

Energy-consistent DMI for varying D

Cell-centred $D_i(\mathrm{m}_{i+1} - \mathrm{m}_{i-1})/2d$ is not the gradient of any discrete energy when D varies. Link form:

$$D\,\partial_x\,\mathrm{m} \to \frac{D_i^{+}\mathrm{m}_{i+1} - D_i^{-}\mathrm{m}_{i-1}}{2d}$$

with $D_i^{\pm} = \frac{1}{2}(D_i + D_{i\pm1})$, or the `inter_dmi` value across region boundaries. Field = exact energy derivative (error 10^{-10}, was 50%).

Demagnetising field and solvers

$$\mathrm{H}_d(\mathrm{r}_i) = -\sum_j N(\mathrm{r}_i - \mathrm{r}_j)\, M_s \mathrm{m}_j$$

Newell cell-averaged tensor (asymptotic far field), six components, FFT convolution with zero padding; periodic axes by macro-geometry image sums. The tensor is symmetric: 6 independent components in Fourier space.

`precision = "single"`: FFTs in float32 as on mumax3 GPUs, 2× faster, $|\Delta H_d|/|H_d| \sim 2 \times 10^{-7}$.

Edge smoothing: sub-cell volume fractions ϕ_i in $M_s \phi_i \mathrm{m}_i$ for the demag field.

Solvers

adaptive Dormand–Prince RK45 (FSAL) and Bogacki–Shampine RK23: $\Delta t' = 0.9\,\Delta t\,(\mathrm{tol}/\mathrm{err})^{1/p}$; fixed-step Heun (thermal), RK4, RK2, Euler; Barzilai–Borwein minimiser; relax; hysteresis.

Moving frame (mumax3 `ext_centerWall`): wall offset $x_w - L/2 = \mathrm{sgn}(m_c^{\mathrm{left}})\langle m_c \rangle L/2$, shift by whole cells, record lab-frame position.

Spin–orbit torque

$$\mathrm{B}_{\mathrm{SOT}} = B_{\mathrm{DL}}\, \mathrm{m} \times \boldsymbol{\sigma} + B_{\mathrm{FL}}\, \boldsymbol{\sigma}, \qquad B_{\mathrm{DL}} = \frac{\hbar\, \theta_{\mathrm{eff}}\, J}{2e M_s\, t}$$

damping-like torque $-\gamma B_{\mathrm{DL}}\, \mathrm{m} \times (\mathrm{m} \times \boldsymbol{\sigma})$ pushes m towards $\boldsymbol{\sigma}$. From first principles: $\theta_{\mathrm{eff}} = 2\sigma_{\mathrm{SH}}/\sigma_{xx}$ (orbital Hall via η_{LS}).

Spin pumping

$$\Delta\alpha = \frac{g \mu_B\, g_r^{\uparrow\downarrow}}{4\pi M_s\, t}, \qquad \mathrm{j}_s = \frac{\hbar}{4\pi} g_r^{\uparrow\downarrow}\, \mathrm{m} \times \dot{\mathrm{m}}$$

Zhang–Li and Slonczewski

$$\mathrm{u} = \frac{P g \mu_B\, \mathrm{J}}{2e M_s (1+\beta^2)}$$

$$\epsilon(\mathrm{m}) = \frac{P \Lambda^2}{(\Lambda^2+1) + (\Lambda^2-1)\, \mathrm{m}\cdot\mathrm{p}}$$

Thiele equation

$$\mathrm{G} \times (\mathrm{v} - \mathrm{u}) + \mathcal{D}\,(\alpha \mathrm{v} - \beta \mathrm{u}) = 0$$

$\mathrm{G} = 4\pi Q\, \hat{z}$, $\mathcal{D}_{ij} = \int \partial_i \mathrm{m} \cdot \partial_j \mathrm{m}\, dA$; predicted velocity and skyrmion Hall angle.

Energy density

$$w = \sum_{s=A,B} w_s[\mathrm{m}_s] - \frac{J_{AF}}{t}\,\mathrm{m}_A\cdot\mathrm{m}_B + A_{AB}\,\nabla\mathrm{m}_A:\nabla\mathrm{m}_B$$

each sublattice with its own M_s, A, D, K, α; demag from $\mathrm{M}_A + \mathrm{M}_B$; one LLG per sublattice.

d-wave altermagnet

$$A_A^x = A_B^y = A(1+\delta), \qquad A_A^y = A_B^x = A(1-\delta)$$

splits the two magnon chiralities along k_x and k_y with opposite sign.

Checks: uniform AFM resonance

$$\omega_\pm = \gamma\left[\sqrt{B_K(2B_E + B_K)} \pm B\right]$$

and the chirality-resolved spin-wave dispersion from the `dispersion` stage; AFM domain-wall velocity under SOT.

LLB equation (Garanin)

$$\dot{\mathrm{m}} = -\gamma \mathrm{m} \times \mathrm{B} + \frac{\gamma \alpha_\parallel}{m^2} (\mathrm{m} \cdot \mathrm{B})\, \mathrm{m} - \frac{\gamma \alpha_\perp}{m^2}\, \mathrm{m} \times (\mathrm{m} \times \mathrm{B})$$

$$\alpha_\parallel = \lambda \frac{2T}{3T_C}, \quad \alpha_\perp = \lambda \Big(1 - \frac{T}{3T_C}\Big) \quad (T < T_C)$$

$$\mathrm{B}_\parallel = \frac{1}{2\chi_\parallel} \Big(1 - \frac{m^2}{m_e^2}\Big) \mathrm{m} \quad (T < T_C)$$

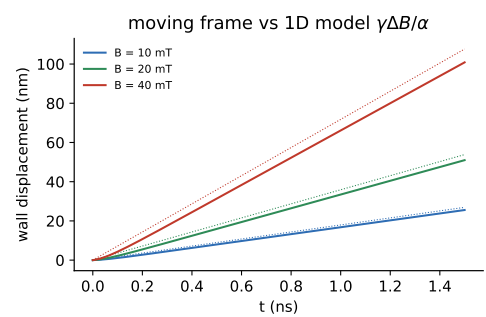
Mean field: $m_e = L(3T_C\, m_e / T)$,
$\chi_\parallel = \frac{\mu_s}{k_B T} \frac{L'(x)}{1 - \frac{3T_C}{T} L'(x)}$.

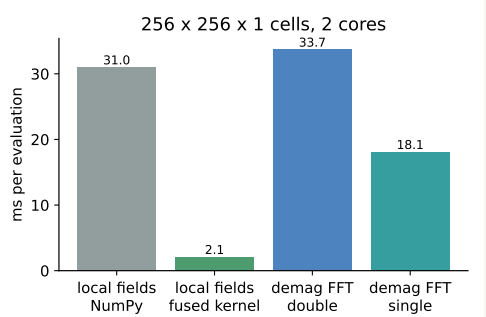
Callen–Callen scaling of the $T = 0$ parameters:

$$M_s \propto m_e, \;\; A, D \propto m_e^{1.8}, \;\; K \propto m_e^3$$

with $m_e(T)$ from mean field or from a magex Monte Carlo run. Two-temperature model for ultrafast demagnetisation.

Micromagnetics: moving frame and kernel speed





Take-away: the moving frame follows the 1D-model wall velocity; the fused local-field kernel is 15× faster than the NumPy stencil, so the demag FFT dominates.

Part VII – Implementation, performance, validation

Multithreaded kernels

Symmetry-reduced couplings. The $3N \times 3N$ CSR matrix becomes a neighbour list; every distinct 3×3 block is stored once (translational invariance) and classified:

$$J_{ij} \in \begin{cases} J\mathbb{1} & \text{1 number} \\ J\mathbb{1} + [\mathrm{D}]_\times & \text{4 numbers} \\ \text{general} & \text{9 numbers} \end{cases}$$

8 bytes of memory traffic per neighbour.

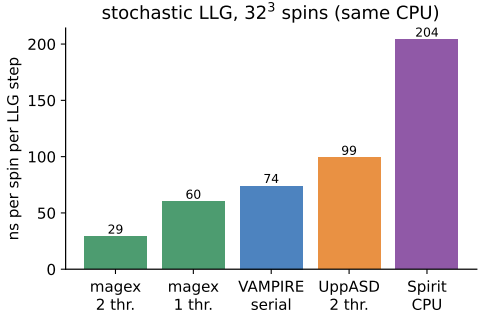
Fused steps. Heun = two passes: (noise + field + predictor), (field at predictor + corrector + norm). Monte Carlo colours swept in parallel; serial twins below 2048 sites.

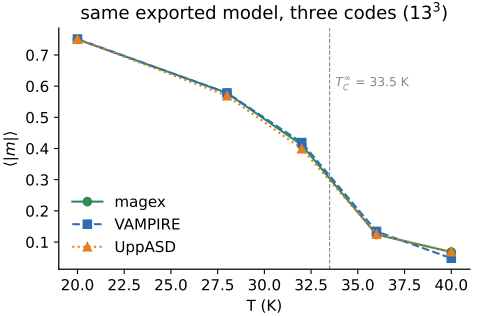
Reproducible parallel random numbers

xorshift64* streams on 256 fixed chunks of the site list (results independent of the number of threads); normals by the Marsaglia–Tsang ziggurat (128 layers): one 32-bit draw and one comparison in 98.8% of calls.

All kernels agree with the NumPy reference to 10^{-15}; MC checked against exact single-spin integrals.

Speed and agreement with VAMPIRE, UppASD and Spirit





All codes built from source on the same 2-core CPU; VAMPIRE serial, UppASD OpenMP, Spirit CPU. MC: magex 80 (1 thread) / 50 (2 threads) ns per spin-sweep vs VAMPIRE 110.

Spirit: same energy for a perturbed skyrmion to 3×10^{-8}; UppASD: same LLG relaxation (0.917 vs 0.918). Production GPU codes (VAMPIRE, Spirit CUDA, mumax3) remain ahead until magex's CUDA kernels are timed on hardware.

Distributed memory: MPI domain decomposition

Slabs along the longest supercell axis. Rank r owns sites $\mathcal{O}_r$ and keeps ghosts $\mathcal{G}_r = \{j \notin \mathcal{O}_r : J_{ij} \neq 0,\ i \in \mathcal{O}_r\}$; local matrix = owned rows only.

$$\underbrace{\text{noise} + \text{field} + \text{predictor}}_{\text{own sites}} \to \text{halo}(\tilde{e}) \to \underbrace{\text{corrector}}_{\text{own}} \to \text{halo}(e)$$

Monte Carlo: every global colour group is split among the ranks, one halo exchange per colour, so the algorithm is exactly the serial one. Non-blocking point-to-point messages; energies and moments by `allreduce`.

131 072 spins	LLG ns/spin/step	MC ns/spin/sweep
1 rank	53.8	71.9
2 ranks	30.4	39.1
speed-up	1.77	1.84

$T = 0$ trajectory with 2 ranks = serial to 2×10^{-15}; set-up replicated on every rank.

CUDA back end

Atomistic (numba.cuda): one thread per spin; device twins of the CPU kernels:

- coupling table (1, 4 or 9 numbers per bond type), on-site tensors, cubic/higher uniaxial terms
- fused Heun/Depondt predictor and corrector kernels; spins stay on the device
- Metropolis / heat-bath colour sweeps, `atomicAdd` acceptance counter
- per-thread xoroshiro128+ random streams

Micromagnetics (CuPy): all arrays on the device; the demag convolution runs through cuFFT, optionally in single precision.

Verification in numba's CUDA simulator:

- $T = 0$ LLG trajectories equal to the CPU kernels (10^{-16})
- Monte Carlo averages equal to the exact single-spin integral
- thermal LLG runs agree with the CPU solver within statistics

No GPU was available: speed on hardware is still to be measured.

Conventions when exchanging models with other codes

code	Hamiltonian	conversion from magex
magex / TB2J	$-\sum_{i\neq j}[J\,\mathrm{e}\cdot\mathrm{e} + \mathrm{D}\cdot(\mathrm{e}\times\mathrm{e}) + \mathrm{e}\cdot J^S\cdot\mathrm{e}]$	meV / eV
UppASD	same, ordered pairs, mRy; $E_K = -K_1 \sin^2\theta$	$J/13.6057$; $K_1 = -K$ (easy axis)
VAMPIRE	tensorial UCF, joules, each bond once	$J_V = 2J \cdot 1.602 \times 10^{-22}$
Spirit	$-\sum_{\langle ij\rangle}[J\,\mathrm{e}\cdot\mathrm{e} + \mathrm{D}\cdot(\mathrm{e}\times\mathrm{e})]$, unique pairs	$J_S = 2J$, $D_S = 2D$ (J^S dropped)
mumax3	continuum (TOML $\rightarrow$ .mx3)	regions, CSG, K_{c1-3}, $B_{1,2}$, solvers

Every keyword the exporters write was checked against the parsers in the VAMPIRE, UppASD and Spirit sources; structure, program and LLG settings are translated too (`create:sphere`, `dipole:solver = atomistic`, `do_dip`, `ddi_method fft`, ...).

Validation summary

check	reference	magex
FFT dipolar field (open, two sublattices)	direct pair sum	10^{-16}
film shape anisotropy (thick film)	$\frac{\mu_0}{2} M_s^2 V$	$< 1\%$
exact dipolar MC	brute-force Metropolis	within 1σ
MC moves, independent spins	exact integral	within 1σ
constrained MC $K(T)$	m^3 (Callen–Callen)	0.5%
LSD $\langle s \rangle$	LSF Monte Carlo	10^{-3}
integrators (Larmor)	analytic	$\mathcal{O}(\Delta t^2)$
Zhang–Li wall velocity	$(\beta/\alpha)u$	2%
FMR	Kittel	0.1%
HTST prefactor (biaxial spin)	f_{FMR}	10^{-12}
sparse HTST (skyrmion)	dense HTST	10^{-8}
MPI LLG, 2 ranks ($T = 0$)	serial trajectory	2×10^{-15}
CUDA kernels (simulator)	CPU kernels	10^{-16}
eigenfrequencies of a FM ring	analytic magnons	10^{-8}
spin–lattice energy error	second order	ratio 4.0 per halving
micromag. terms (Kc2, Kc3, B_1, B_2, inter-DMI, custom)	finite differences	10^{-9}
moving frame wall velocity	$\gamma \Delta B / \alpha$	3%
m(T), simple cubic	VAMPIRE, UppASD	within critical fluctuations
energy of a skyrmion configuration	Spirit	3×10^{-8}

Using the engine

```
[spinmodel]
results   = "magex_results/results.json"
supercell = [14, 14, 14]
pbc       = [false, false, false]

[structure]
shape      = "sphere"
size       = 20.0
core_shell = {radius = 12.0, core = "core",
              shell = "shell"}
materials  = [{name = "core", Kc = 0.02},
              {name = "shell", ks = 0.05}]
exchange_scale = [[1,1,1],[1,1,-0.3],[1,-0.3,0.6]]
dipolar    = true

[program]
type    = "field_cooling"
T_start = 60.0
field   = [0, 0, 1.0]
```

```
magex run      input.toml   # exchange (Part I)
magex llg      llg.toml     # spin dynamics
magex mc       mc.toml      # T_C, PT, WL
magex program  prog.toml    # hysteresis, FMR,
                            # CMC, LSF, ...
magex gneb     gneb.toml    # barriers + HTST
magex sqw      sqw.toml     # S(q, w)
magex micromag mm.toml      # continuum
magex export   prog.toml --to vampire
               # | uppasd | spirit | mumax3
```

Output: CSV, NPZ, VTK (ParaView), extended XYZ (OVITO), OVF 2.0, PDF/PNG figures.

Summary

Parameters

- exact force-theorem Hessians, rotational invariance
- J, D, J^{ani}, A from one DFT run
- Bruno, downfolding, Lloyd, LSWT, damping

Simulation

- atomistic LLG, MC, GNEB/HTST, SLD, LSD; MPI and CUDA
- nanostructures, exact dipoles
- mumax3-class micromagnetics, LLB, AFM

Quality

- every formula tested (90+ automated tests)
- same results as VAMPIRE, UppASD, Spirit
- faster per core than all three on CPU

Next: timings of the CUDA back end on GPU hardware; distributed system set-up for 10^8-spin MPI runs; GPU dipolar fields; many-body (EAM) lattice potentials.